

Fenótipos Complexos e Função: Superfícies e Paisagens de Desempenho

‘Evolução de Fenótipos Complexos’

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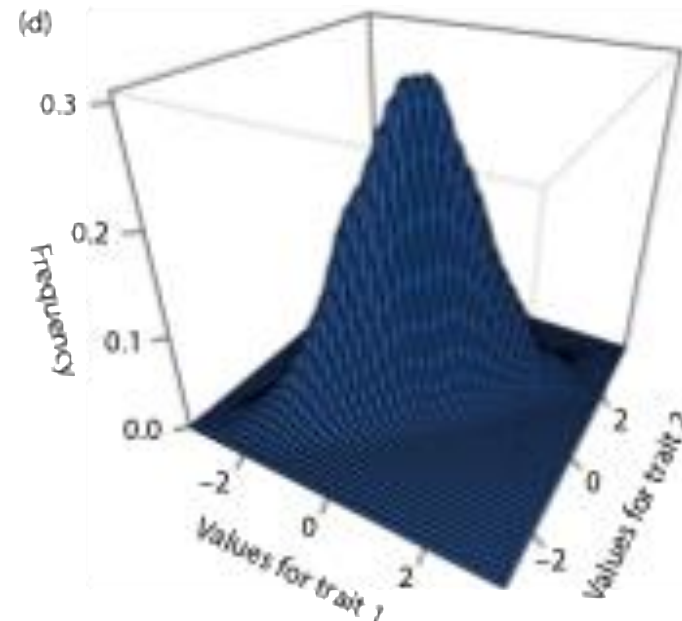
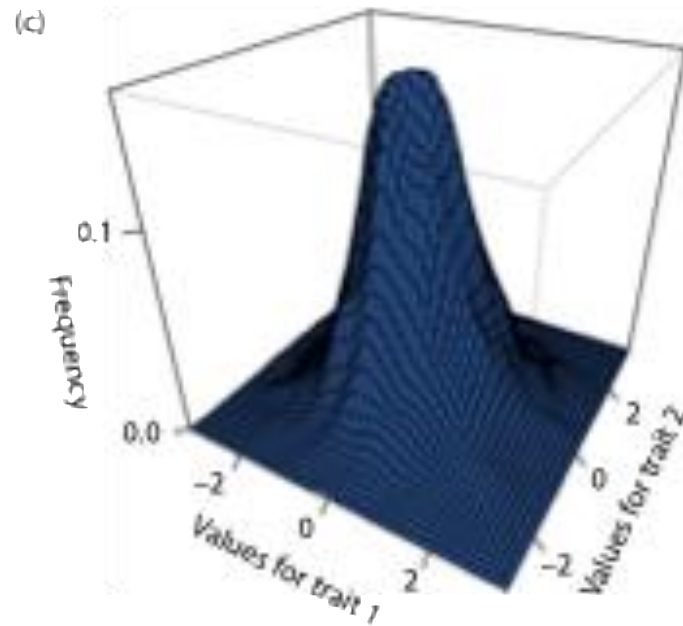
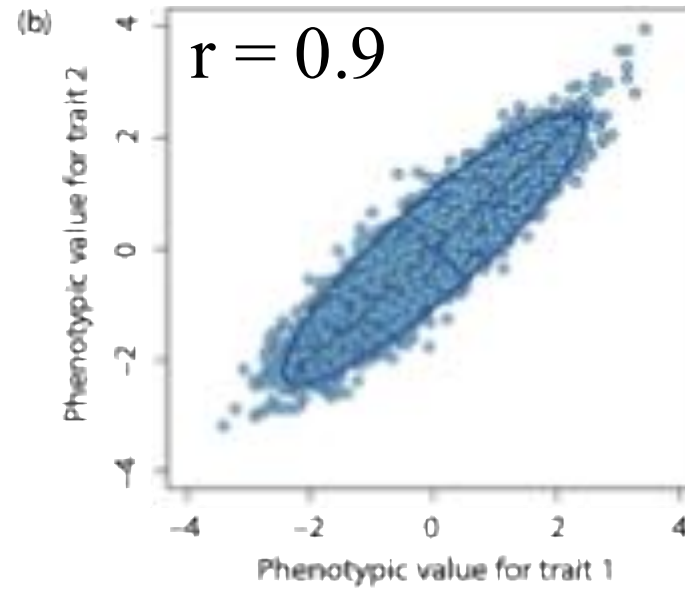
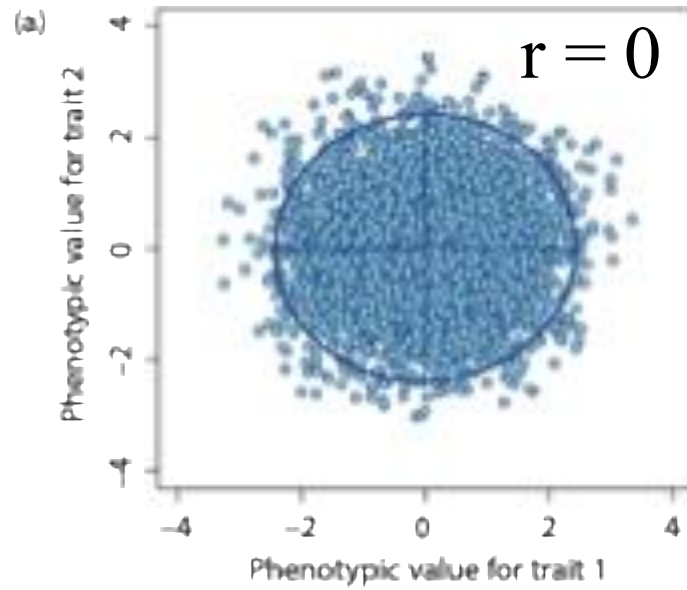
Por que estudar variação em biologia?

Quais as origens de variação fenotípica e genética?

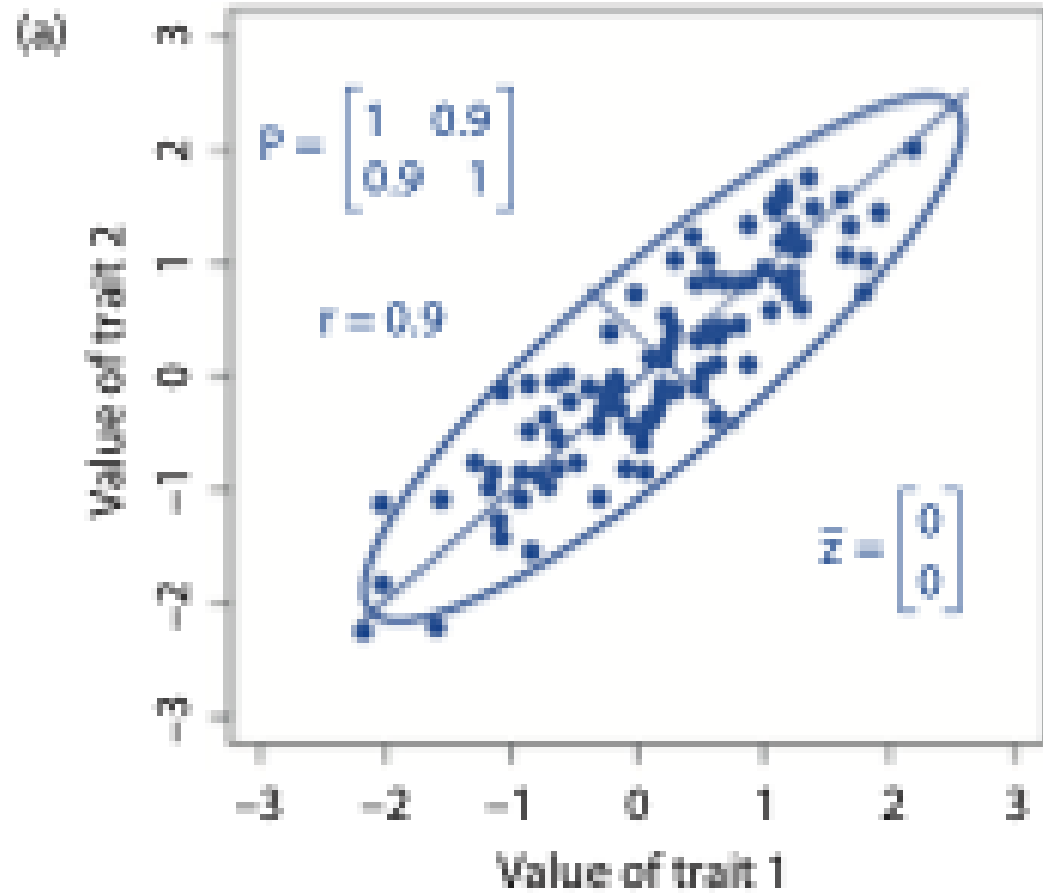
Como a variação é organizada e representada em
populações?

Quais são as consequências evolutivas de como a
variação é organizada nas populações?

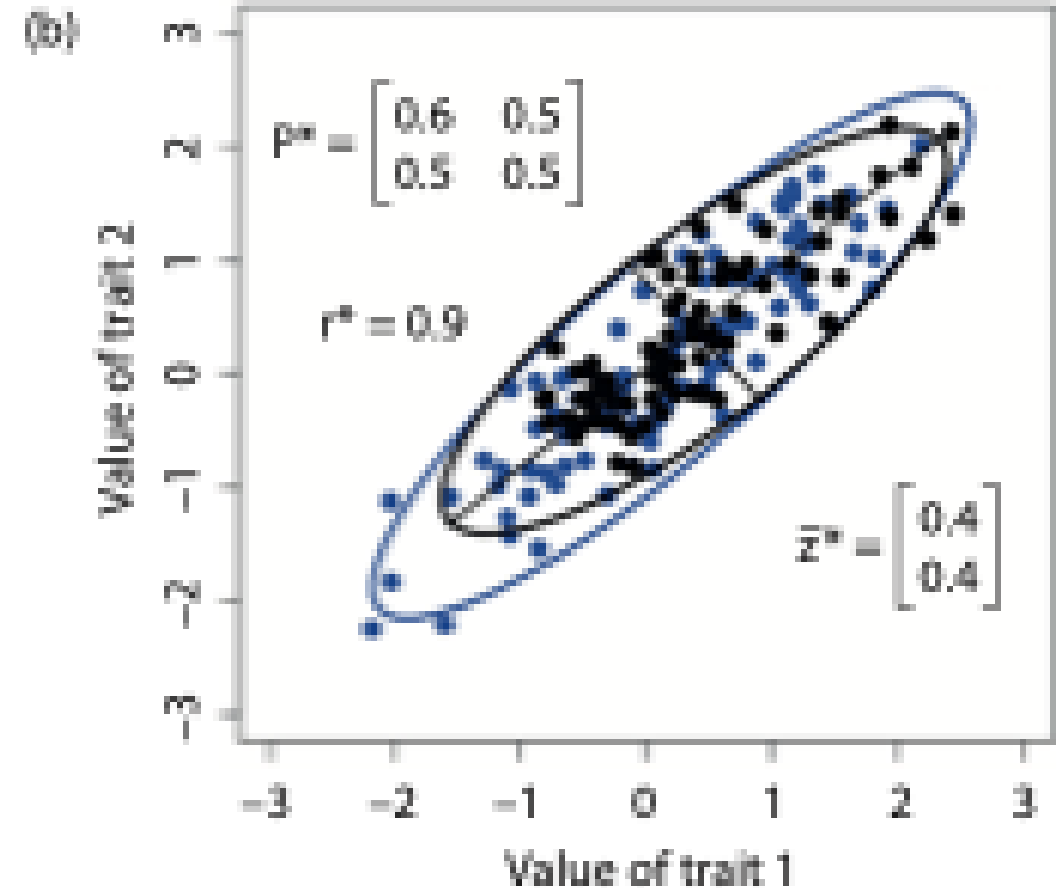
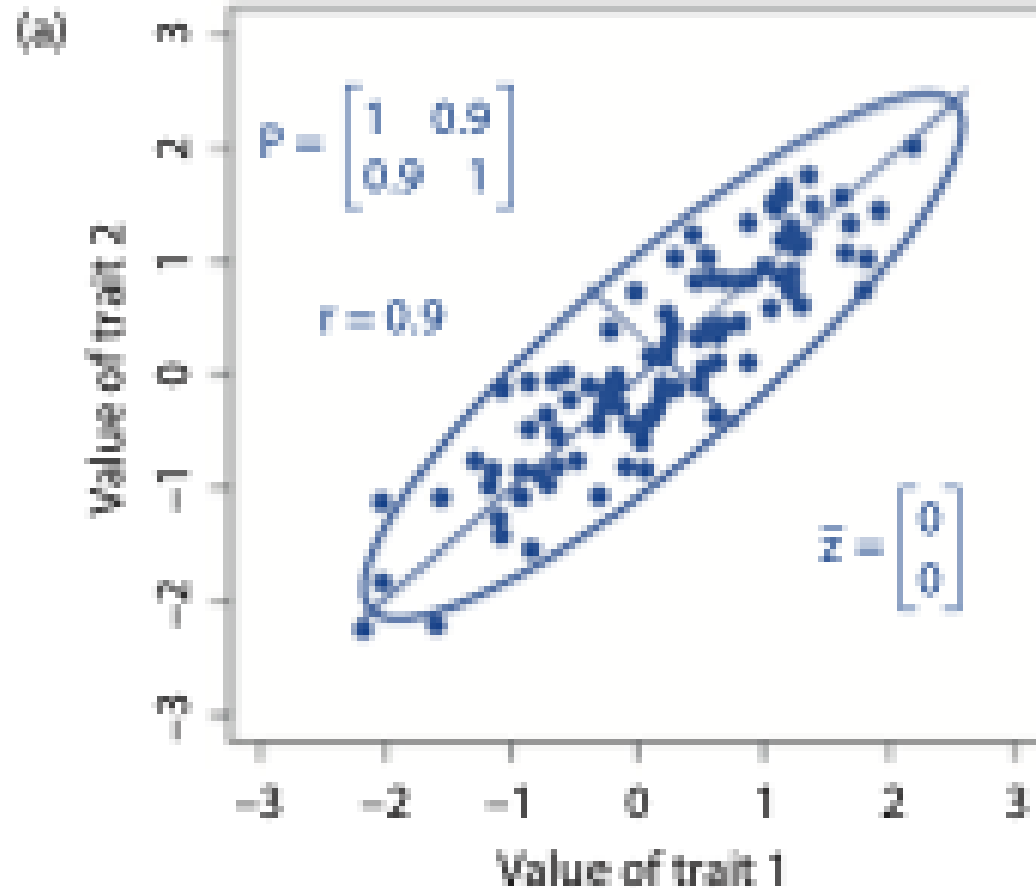
‘Enxergando’ o mundo multivariado



Seleção multivariada



Seleção multivariada



Seleção de truncamento bivariada: $z_1 > -1$ e $z_2 > -1$

Seleção direcional bivariada pode mudar variâncias e covariâncias

Diferencial de seleção multivariado

$$\boxed{s = \text{Cov}(w, z)} = \begin{bmatrix} \text{Cov}(w, z_1) \\ \text{Cov}(w, z_2) \end{bmatrix} = \bar{z} - \bar{z}^* = \begin{bmatrix} \bar{z}_1 - \bar{z}_1^* \\ \bar{z}_2 - \bar{z}_2^* \end{bmatrix} = \begin{bmatrix} s_1 \\ s_2 \end{bmatrix}$$

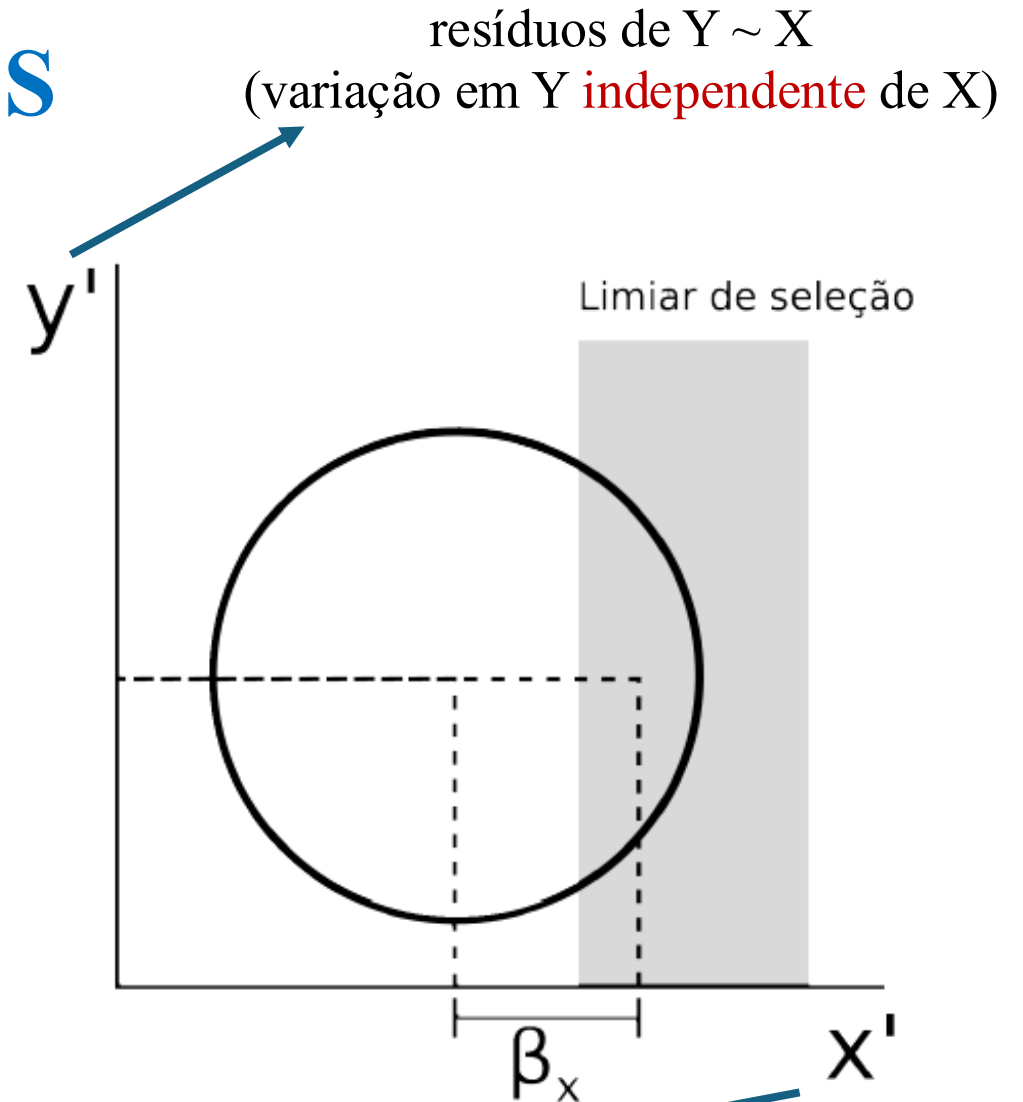
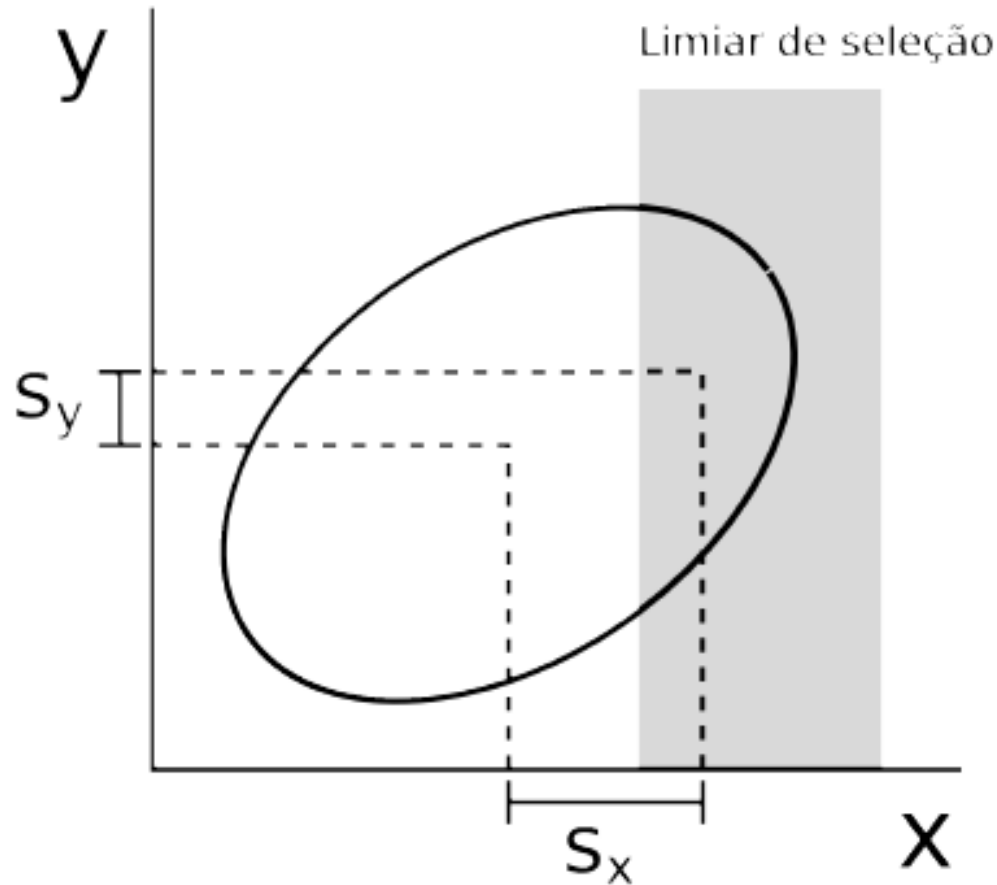
Aptidão relativa é o que traduz a distribuição antes de seleção para a distribuição após seleção (**Equação de Price**).

Gradiente de seleção multivariado

$$\beta \equiv P^{-1}s = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}$$

Gradiente de seleção multivariado

$$\beta = P^{-1} S$$

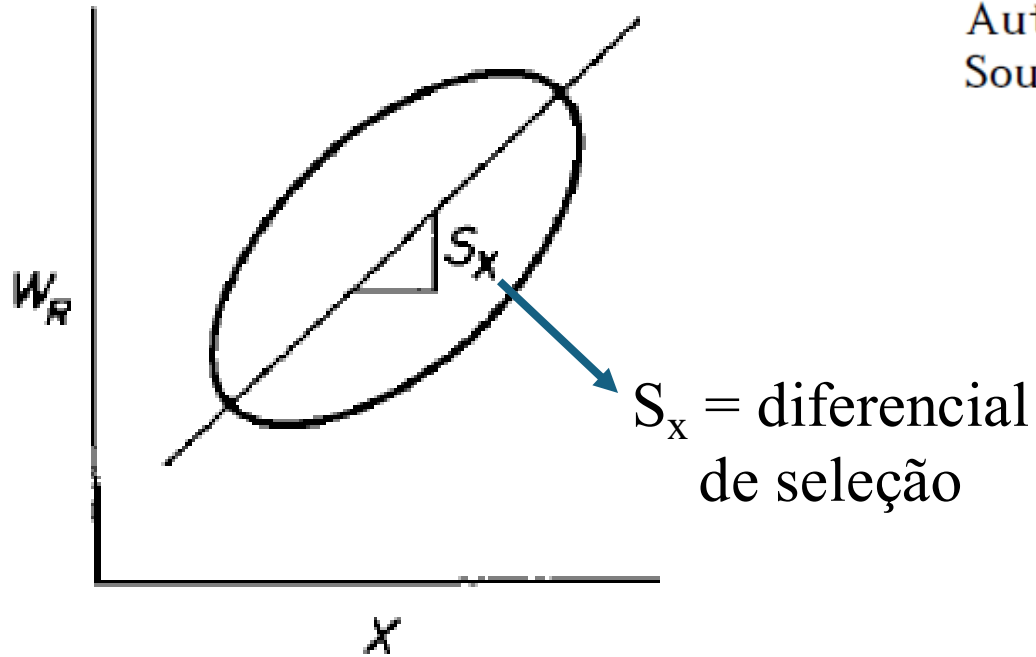


resíduos de $Y \sim X$
(variação em Y independente de X)

resíduos de $X \sim Y$
(variação em X independente de Y)

Relação entre gradiente de seleção e regressão

The Measurement of Selection on Correlated Characters
Author(s): Russell Lande and Stevan J. Arnold
Source: *Evolution*, Vol. 37, No. 6 (Nov., 1983), pp. 1210-1226



$$S_x = \text{cov}(W, x)$$

$$\text{cov}(W, x) = \beta_x \text{var}(x)$$

$$\beta_x = \text{cov}(W, x) / \text{var}(x) \longrightarrow \text{Coeficiente de regressão}$$

$$\beta = P^{-1} S$$

$$\beta = P^{-1} \text{cov}(W, \mathbf{z})$$

Corresponde a um **coeficiente de regressão parcial** (efeito de um caráter ao manter todos os demais fixos = **remover as covariâncias fenotípicas**)

Seleção Indireta

$$s = P\beta = \begin{bmatrix} P_{11} & P_{12} \\ P_{12} & P_{22} \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} = \begin{bmatrix} P_{11}\beta_1 + P_{12}\beta_1 \\ P_{12}\beta_2 + P_{22}\beta_2 \end{bmatrix} = \begin{bmatrix} s_1 \\ s_2 \end{bmatrix}$$

Seleção indireta vem de **covariâncias FENOTÍPICAS** entre os caracteres que estão sobre seleção direta com os demais caracteres.

Gradiente de seleção revela os alvos de seleção!

Diferencial de seleção multivariado não-linear

$$\mathbf{C} \begin{bmatrix} \text{Cov}(w, z_1^2) & \text{Cov}(w, z_1 z_2) \\ \text{Cov}(w, z_1 z_2) & \text{Cov}(w, z_2^2) \end{bmatrix}$$

Covariância entre
aptidão e desvios
cruzados da média

SELEÇÃO
CORRELACIONADA

Covariância entre
aptidão e desvios ao
quadrado da média

Seleção
estabilizadora/disruptiva

Gradiente de seleção multivariado não-linear

$$\mathbf{y} \equiv \mathbf{P}^{-1} \mathbf{C} \mathbf{P}^{-1} = \begin{bmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{12} & \gamma_{22} \end{bmatrix},$$

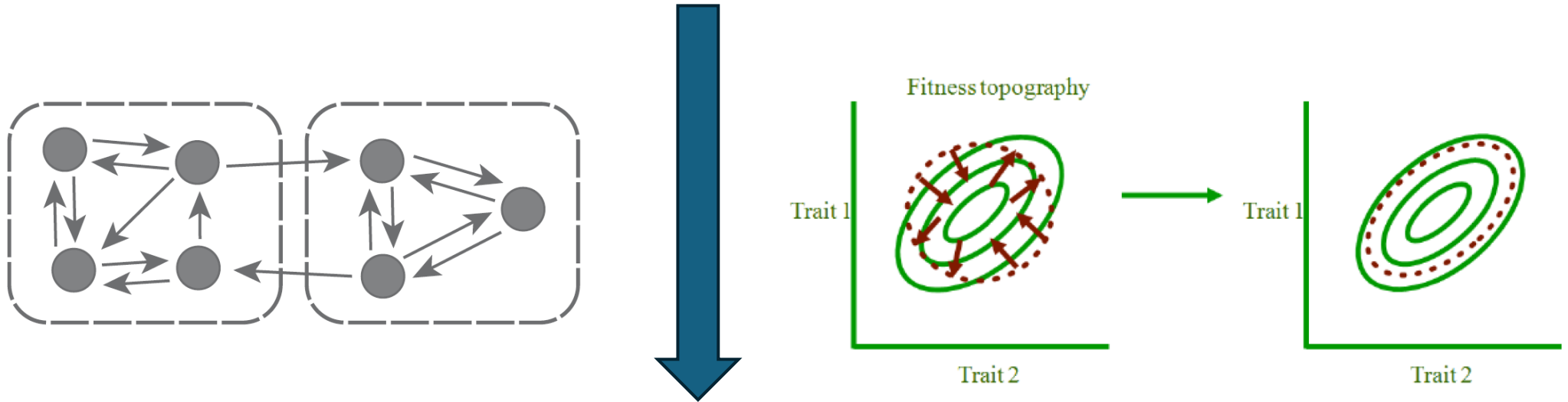
SELEÇÃO
CORRELACIONADA

$$\begin{aligned} \mathbf{C} = \mathbf{P} \mathbf{y} \mathbf{P} &= \begin{bmatrix} P_{11} & P_{12} \\ P_{12} & P_{22} \end{bmatrix} \begin{bmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{12} & \gamma_{22} \end{bmatrix} \begin{bmatrix} P_{11} & P_{12} \\ P_{12} & P_{22} \end{bmatrix} \\ &= \begin{bmatrix} P_{11}^2 \gamma_{11} + 2P_{11}P_{12}\gamma_{12} + P_{12}^2 \gamma_{22} & P_{11}P_{12}\gamma_{11} + P_{12}^2 \gamma_{12} + P_{11}P_{22}\gamma_{12} + P_{12}P_{22}\gamma_{22} \\ P_{11}P_{12}\gamma_{11} + P_{12}^2 \gamma_{12} + P_{11}P_{22}\gamma_{12} + P_{12}P_{22}\gamma_{22} & P_{12}^2 \gamma_{11} + 2P_{12}P_{22}\gamma_{12} + P_{22}^2 \gamma_{22} \end{bmatrix} \end{aligned}$$

Seleção não-linear indireta vem de **covariâncias FENOTÍPICAS** entre os **caracteres ao quadrado**

Seleção Correlacionada

Estimativas de seleção correlacionada podem prover evidência de que **interações entre caracteres** afetam a aptidão média populacional



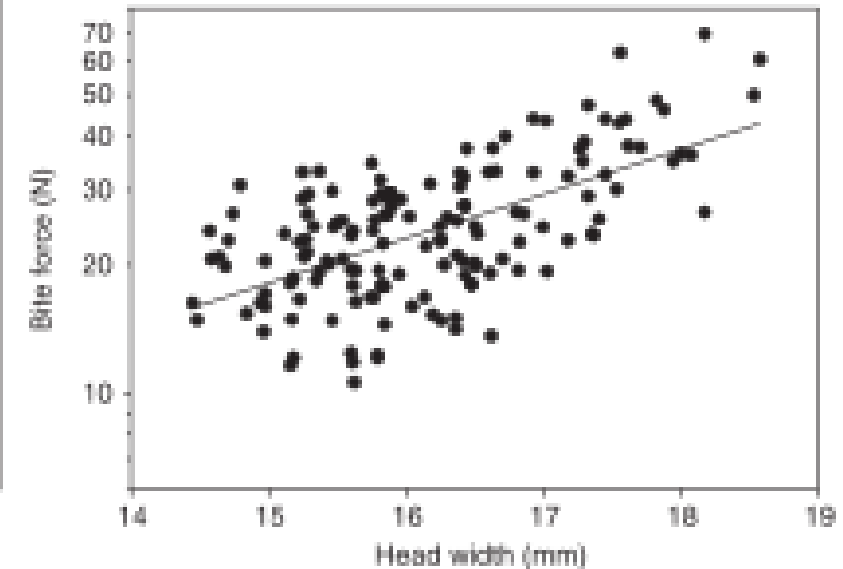
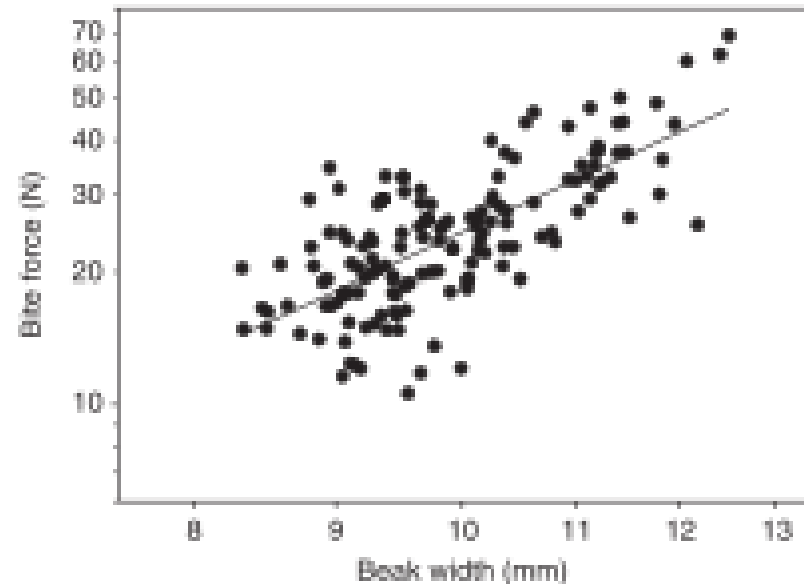
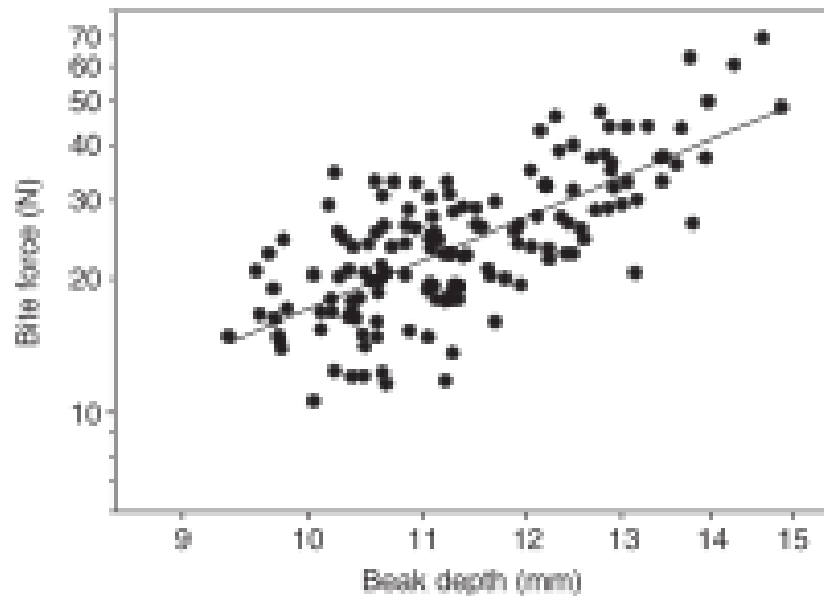
Esperamos que seleção correlacionada molde **COMPLEXOS FUNCIONAIS** que afetam **DESEMPENHO!**

Bite performance and morphology in a population of Darwin's finches: implications for the evolution of beak shape

A. HERREL,*† J. PODOS,‡§ S. K. HUBER§ and A. P. HENDRY¶



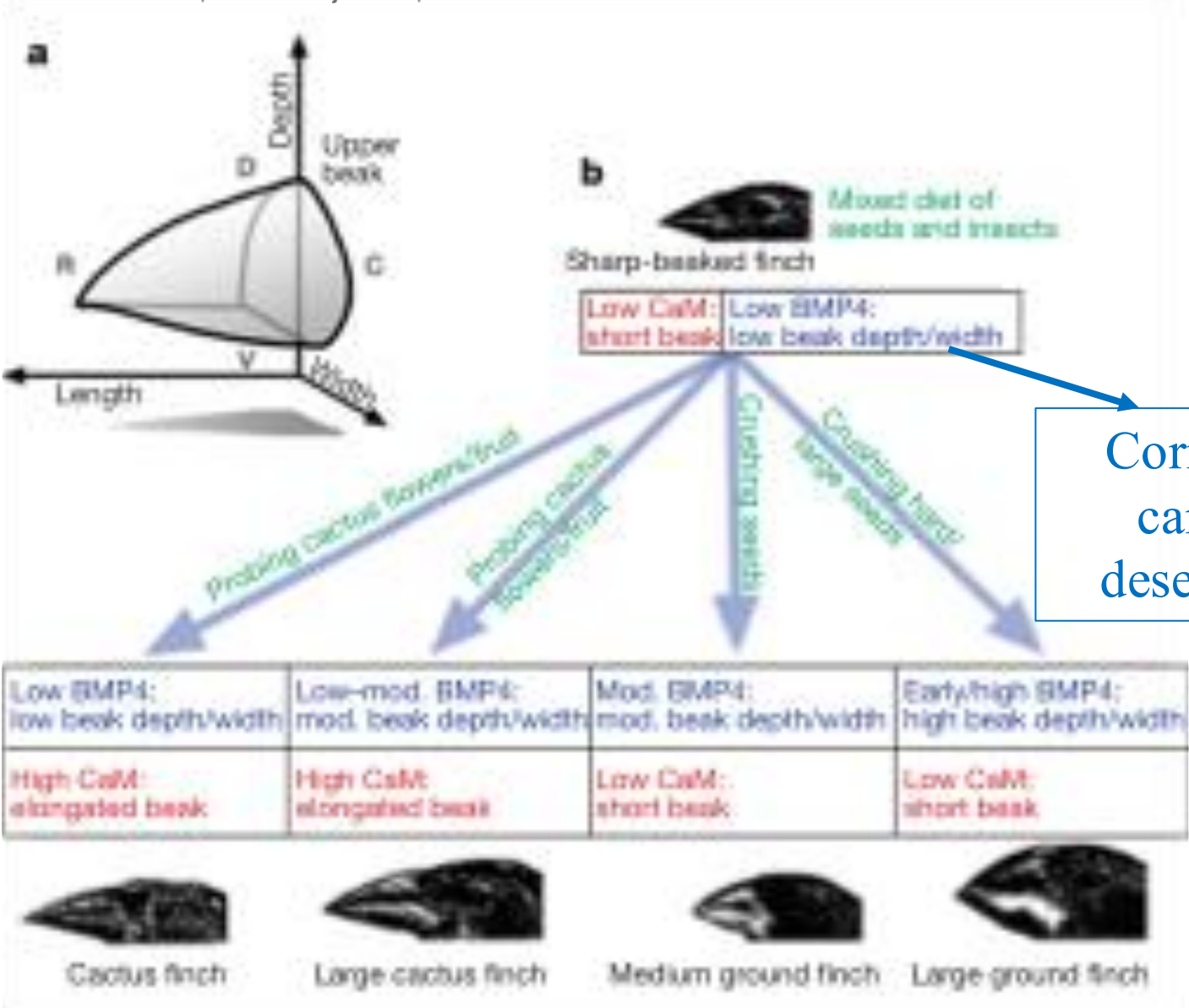
Geospiza fortis



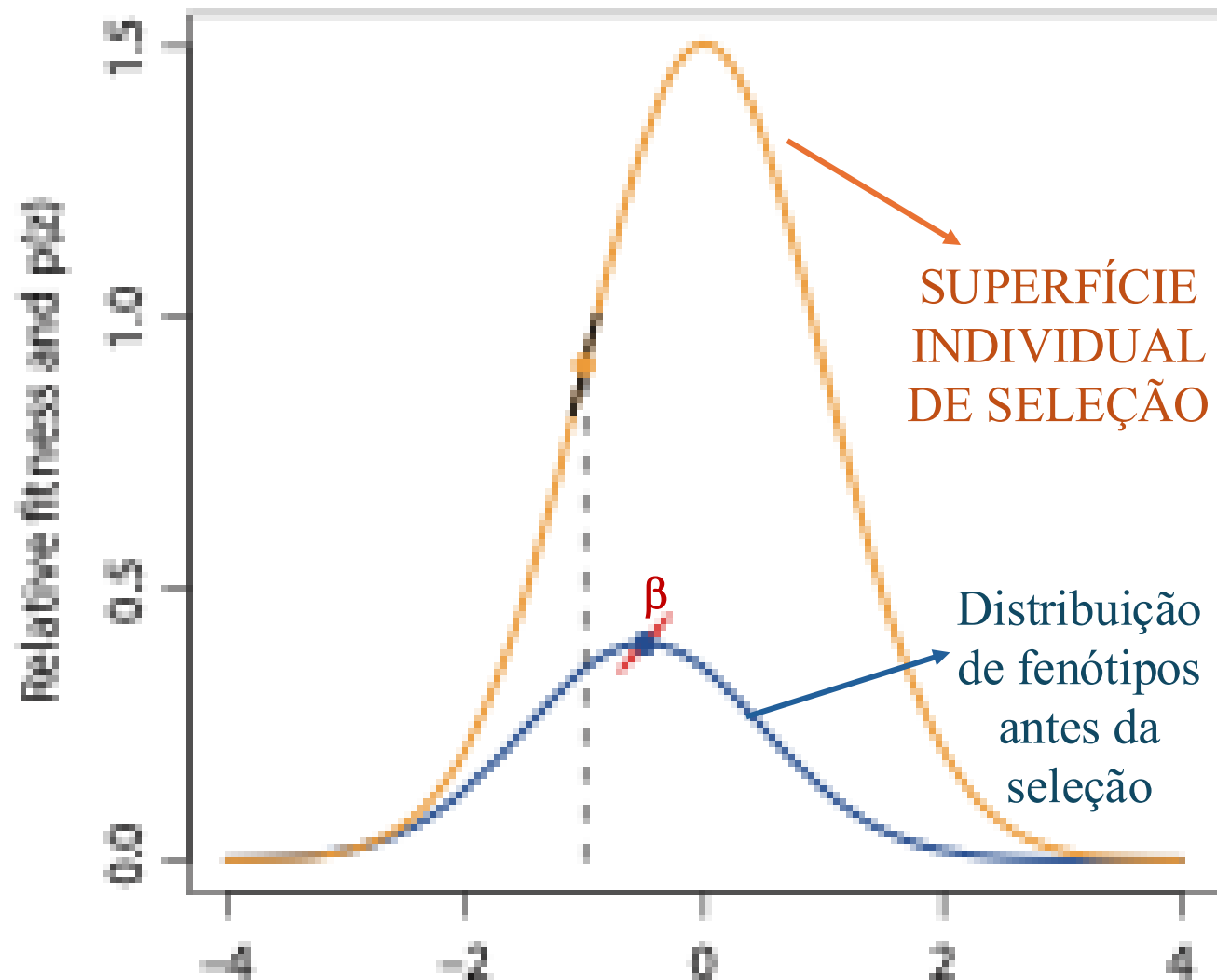
Complexo funcional: profundidade e largura do bico, e largura da cabeça

The calmodulin pathway and evolution of elongated beak morphology in Darwin's finches

Arhat Abzhanov^{1†}, Winston P. Kuo^{1,2,3†}, Christine Hartmann⁴, B. Rosemary Grant⁵, Peter R. Grant⁵ & Clifford J. Tabin¹



‘Enxergando’ seleção linear univariada



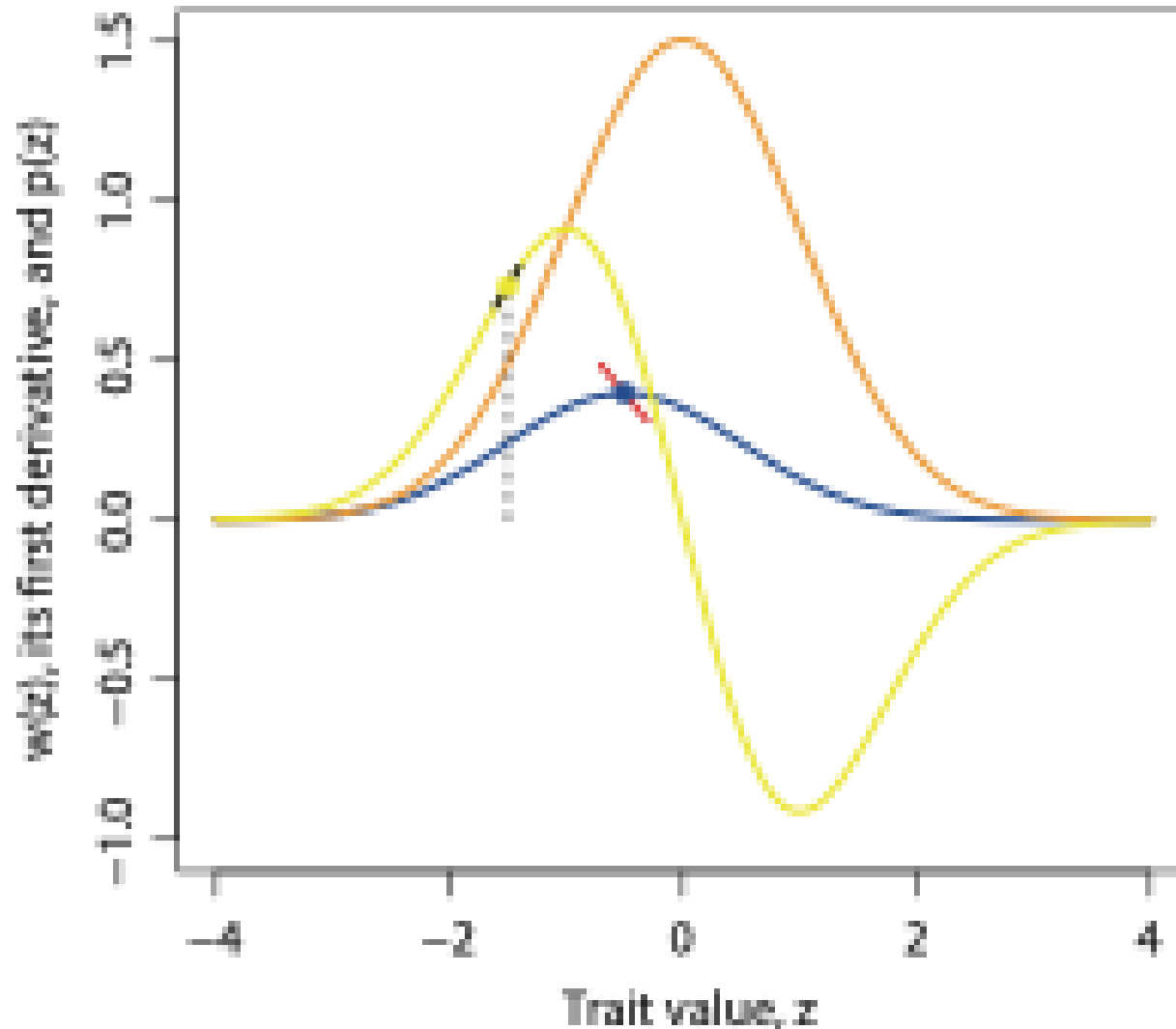
Trait value, z → Desvios padronizados da média populacional

ISS = aptidão relativa dos indivíduos em função de seus valores de z

Inclinação da ISS = ‘força’ de seleção para aquele valor de z = primeira derivada da ISS

Inclinação média da ISS = gradiente linear de seleção!

‘Enxergando’ seleção não-linear univariada



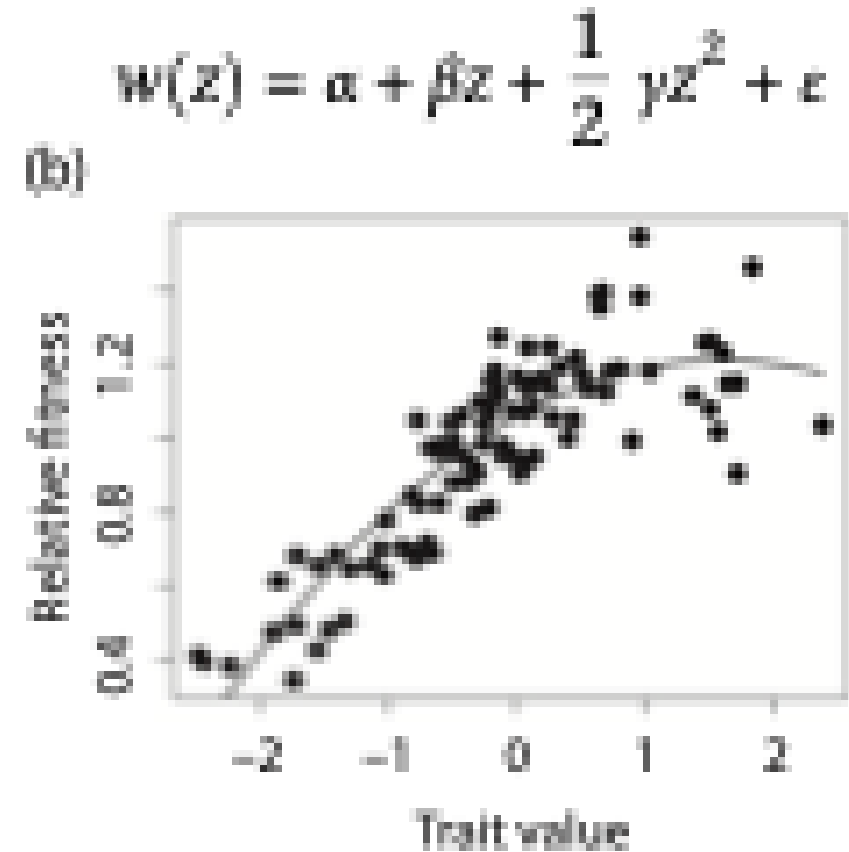
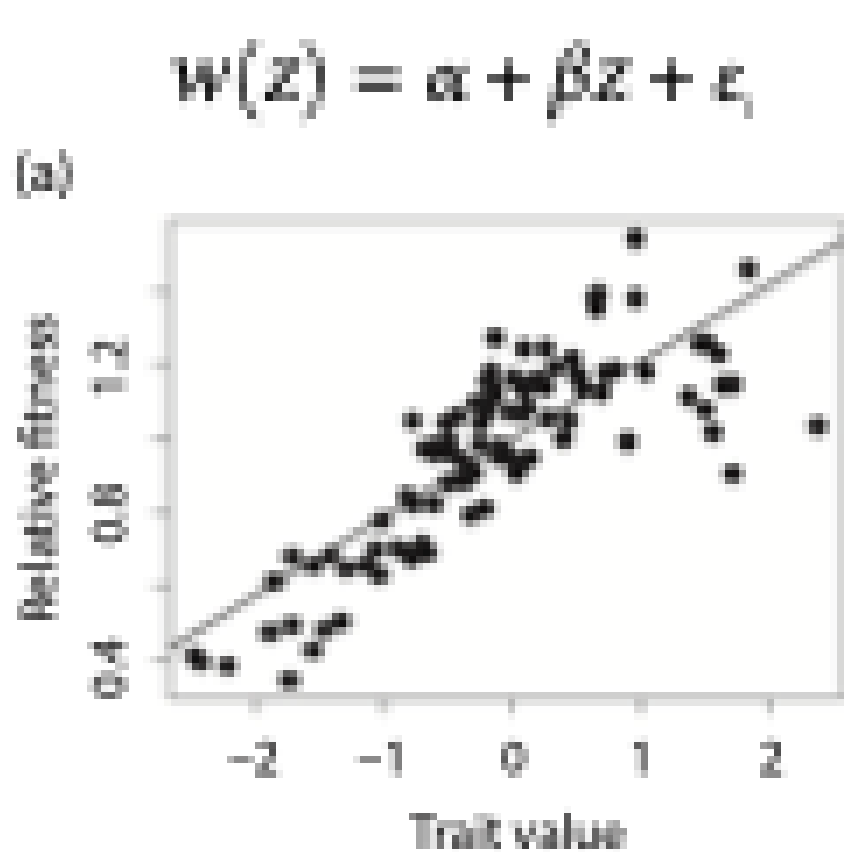
ISS = aptidão relativa dos indivíduos em função de seus valores de z

Curvatura da ISS = ‘força’ de seleção para aquele valor de z^2 = segunda derivada da ISS

Curvatura média da ISS = gradiente não-linear de seleção!

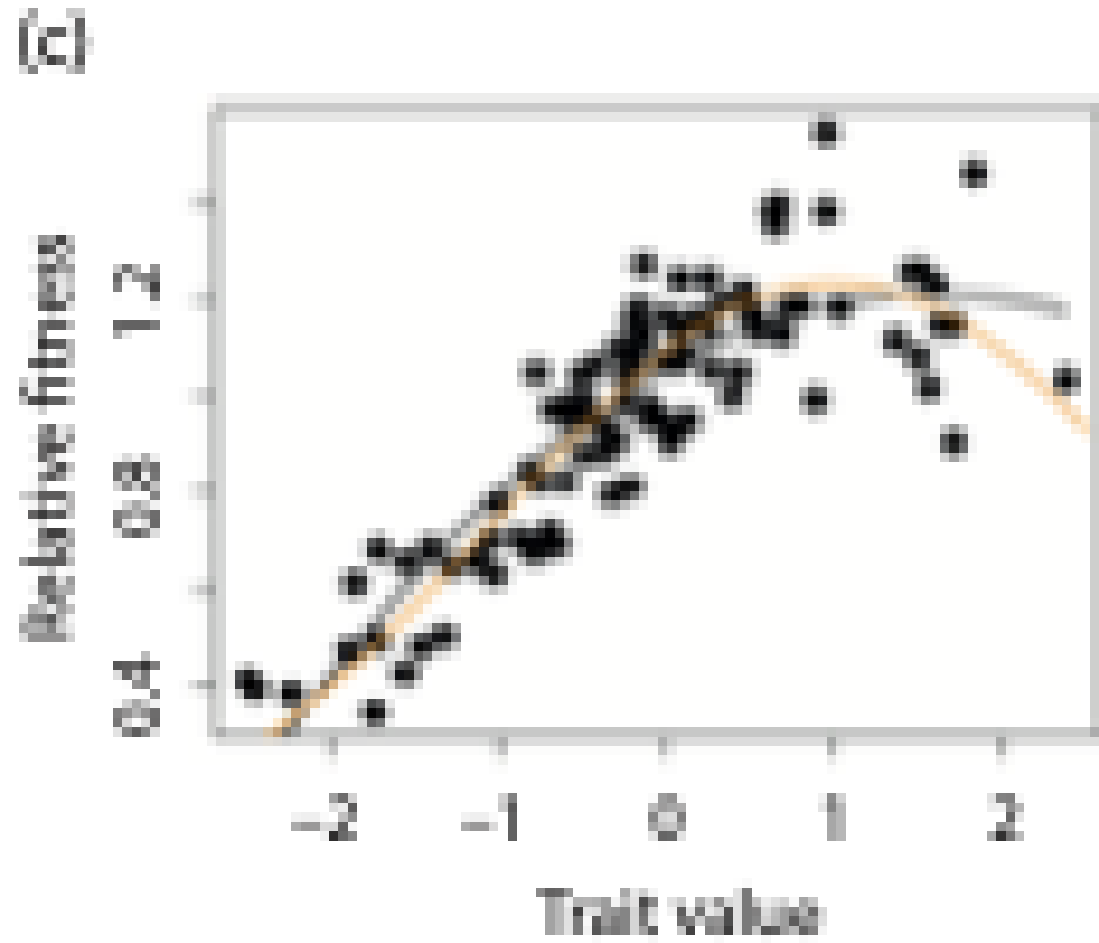
$$\gamma = -0.4375$$

Estimativas de seleção: modelos lineares/quadráticos



Regressão linear e quadrática são **aproximações** que nos permitem estimar os **parâmetros de seleção**.

Função Gaussiana e ISS ‘verdadeira’



ÓTIMO OU PICO
ADAPTATIVO!

$$w(z) = \exp \left\{ \frac{-(z - \theta)^2}{2\sigma^2} \right\},$$

abertura da função

‘Enxergando’ seleção não-linear bivariada

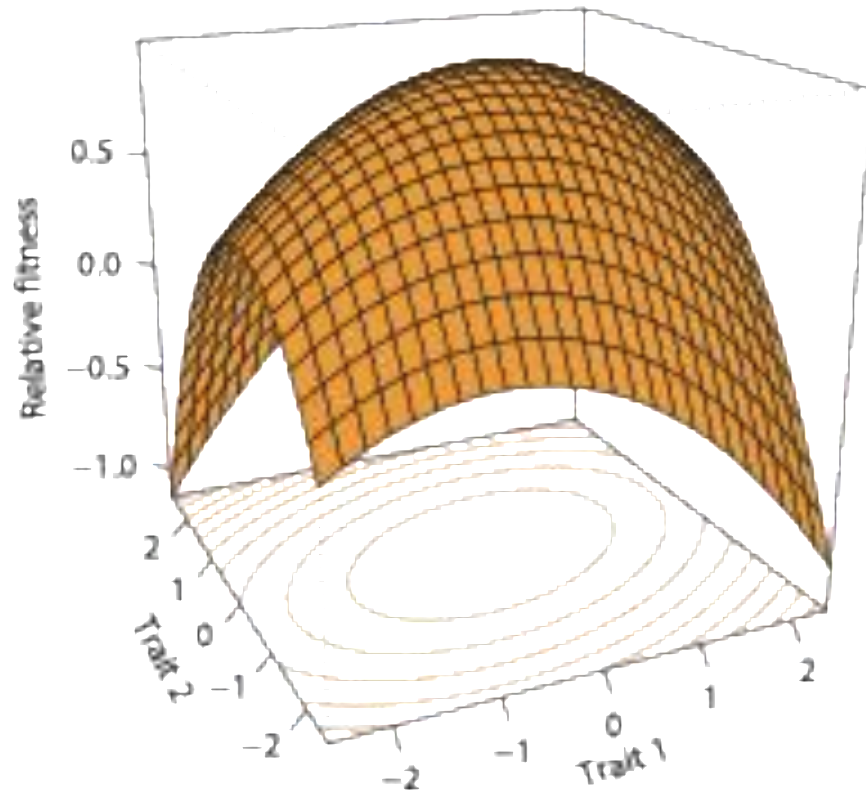
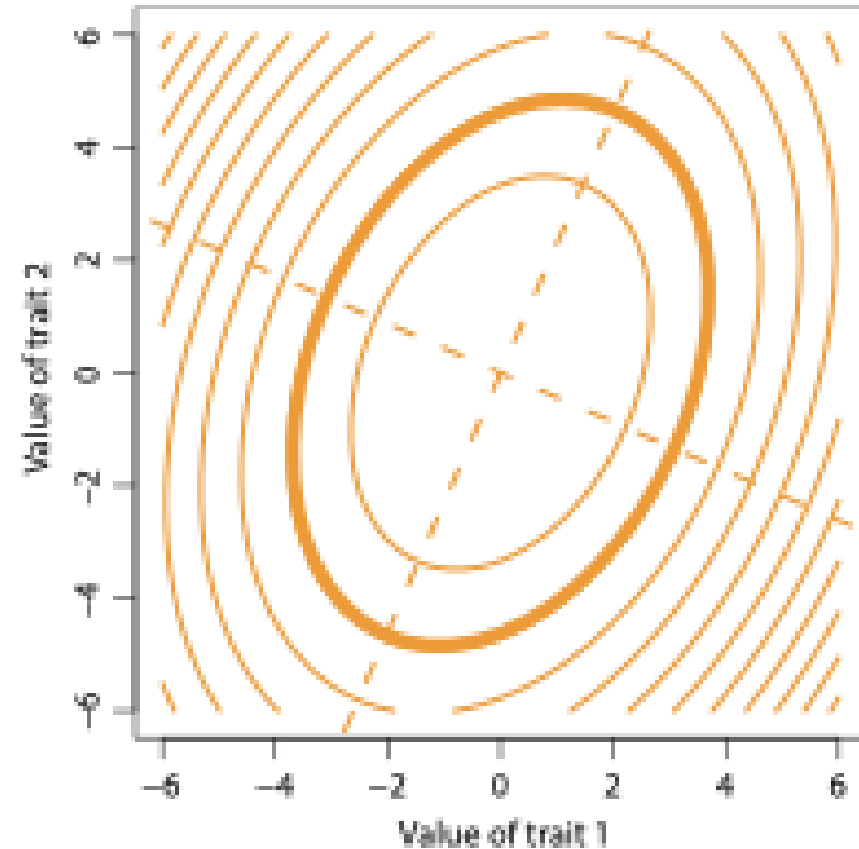


Figure 4.1 Bivariate stabilizing selection represented as a convex selection surface.



$$\alpha = 1, \beta_1 = 0.22, \beta_2 = -0.08, \gamma_{11} = -0.31, \gamma_{22} = -0.18, \text{ and } \gamma_{12} = 0.07$$

Estimativas de seleção não-linear multivariada

$$w(x) = \alpha + \beta x + \frac{1}{2} \gamma x^2 + \epsilon$$



$$w(\mathbf{x}) = \alpha + \beta^T \mathbf{x} + \frac{1}{2} \mathbf{x}^T \gamma \mathbf{x} + \epsilon = \alpha + \beta_1 x_1 + \beta_2 x_2 + \underbrace{\frac{1}{2} \gamma_{11} x_1^2 + \frac{1}{2} \gamma_{22} x_2^2 + \gamma_{12} x_1 x_2}_{\text{Gradients de seleção estabilizadora/disruptiva (variâncias) e de seleção correlacionada (covariâncias)}} + \epsilon,$$

Gradients de seleção
estabilizadora/disruptiva (variâncias) e
de seleção correlacionada
(covariâncias)

Seleção não-linear multivariada em complexos funcionais: **Superfícies de Desempenho**

Bite performance surfaces of three ecologically divergent Iguanidae lizards: relationships with lower jaw bones

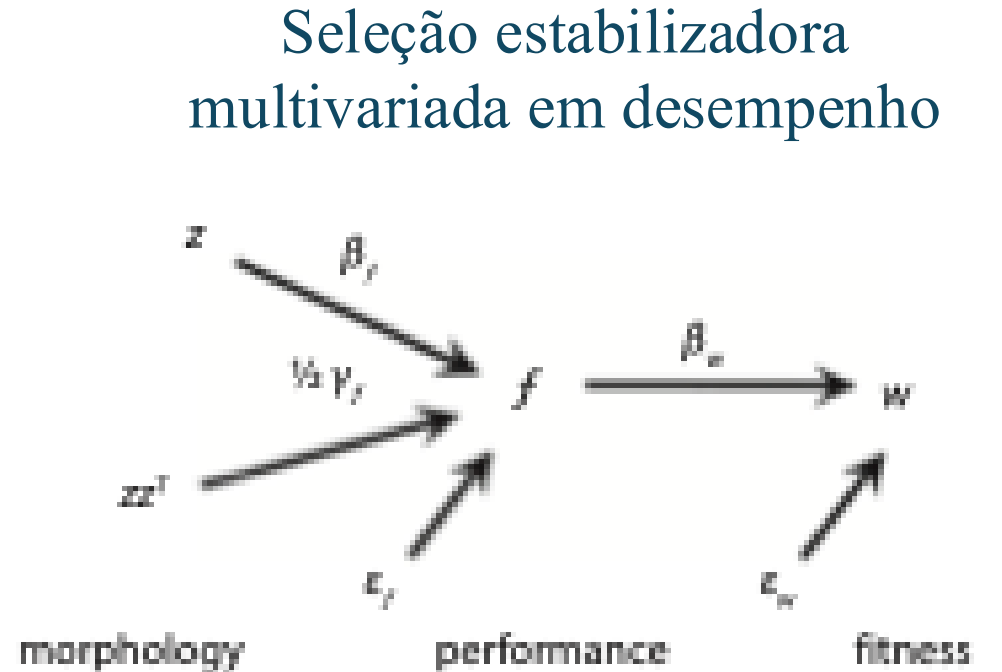
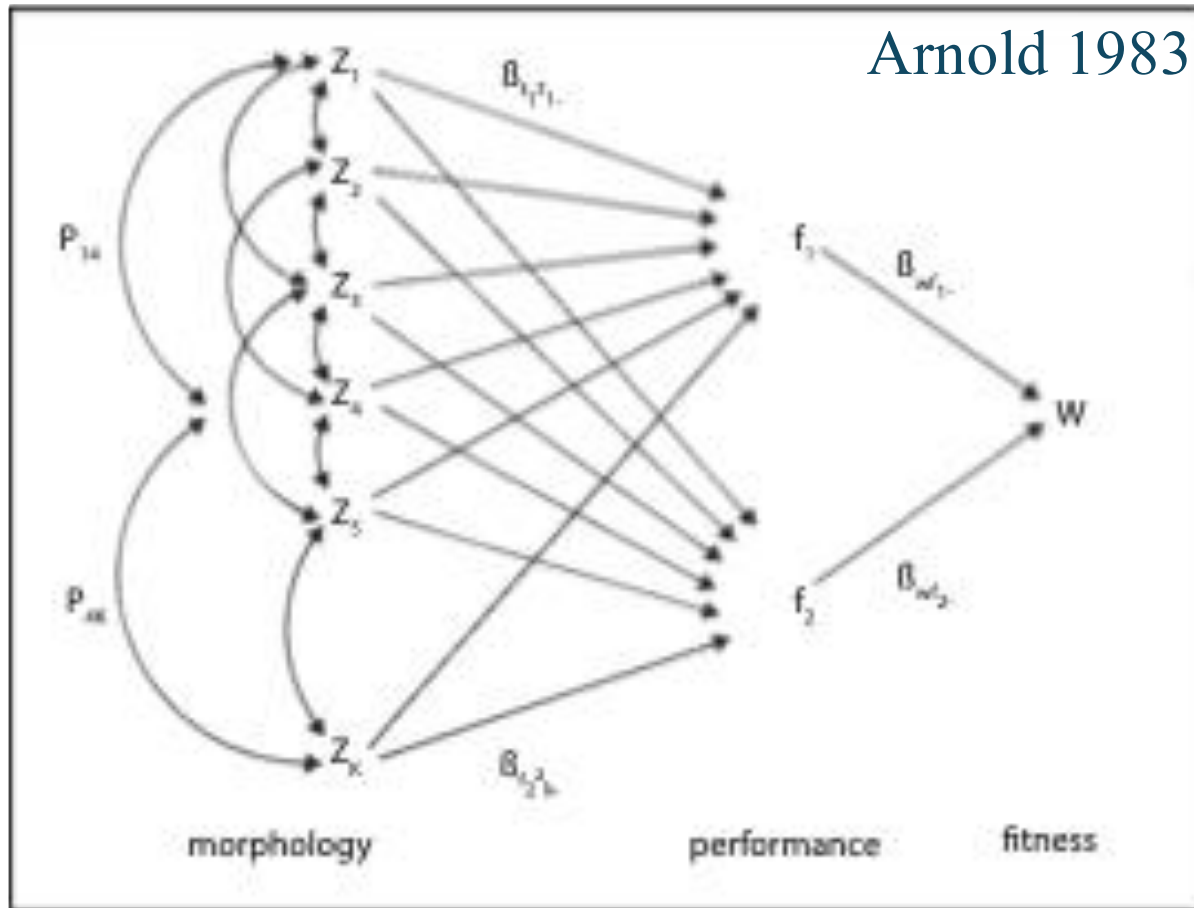
MONIQUE NOUAILHETAS SIMON^{1,2,*}, RENATA BRANDT³, TIANA KOHLSDORF³ and STEVAN J. ARNOLD²

Biological Journal of the Linnean Society, 2019,

‘To study the operation and evolution of functional complexes, however, we need to measure the **successful operation of the complex**. That measure of operation is called **performance**’.

Arnold 2023

Paradigma (linear) do Desempenho



Arnold 2003

Figure 7.5 Path diagram showing causal relationships (paths with single arrowhead) between k morphological traits ($z_1, z_2, \dots, z_k$), two measures of performance (f_1 and f_2), and relative fitness (w).

Lagartos e Força de mordida

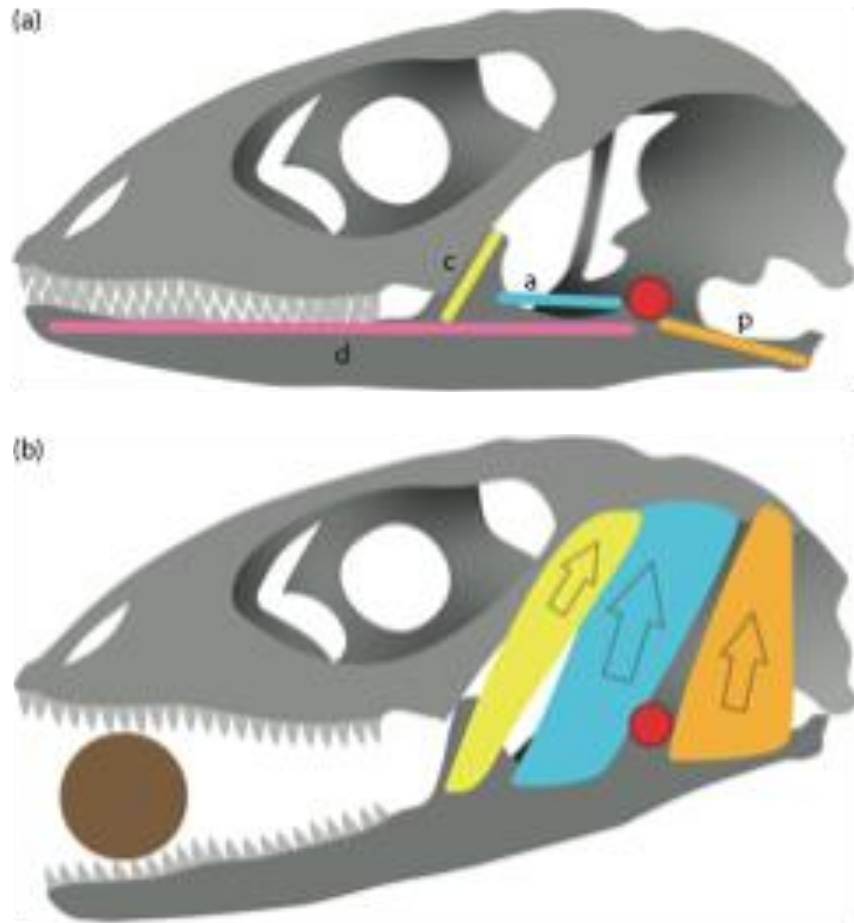
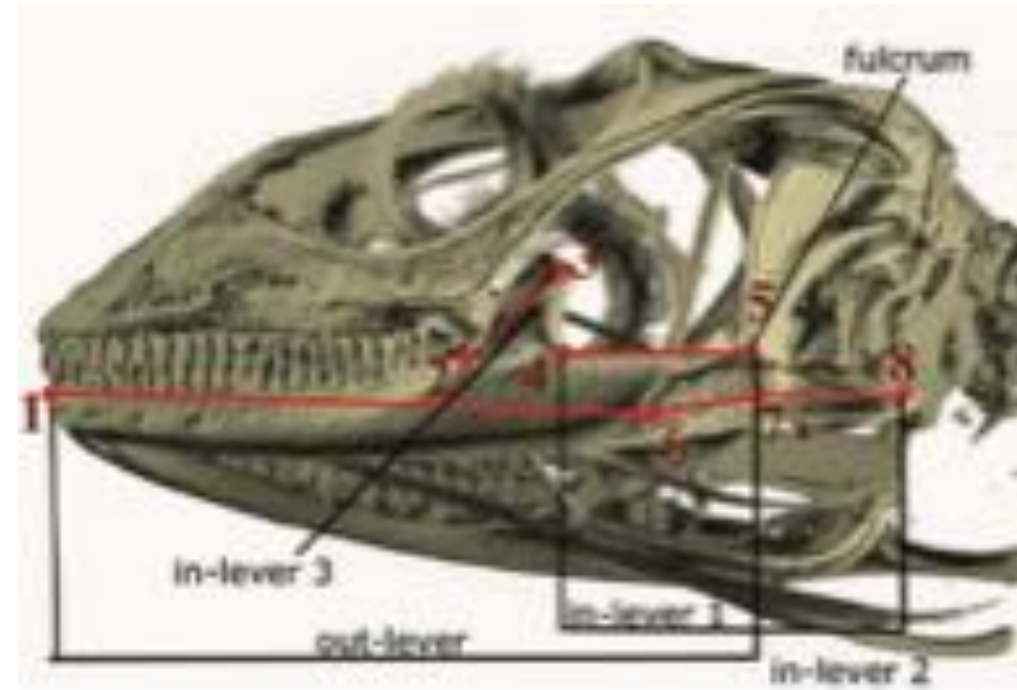


Figure 7.9 Biomechanical models for the bite force of the lower jaw of the lizard *Tropidurus hispidus*. (a) The diagram



Modelo biomecânico para o aumento de força de mordida:

$$\text{VANTAGEM MECÂNICA} = \frac{\text{In-lever}}{\text{Out-lever}}$$

Previsão de onde e como será a seleção não-linear

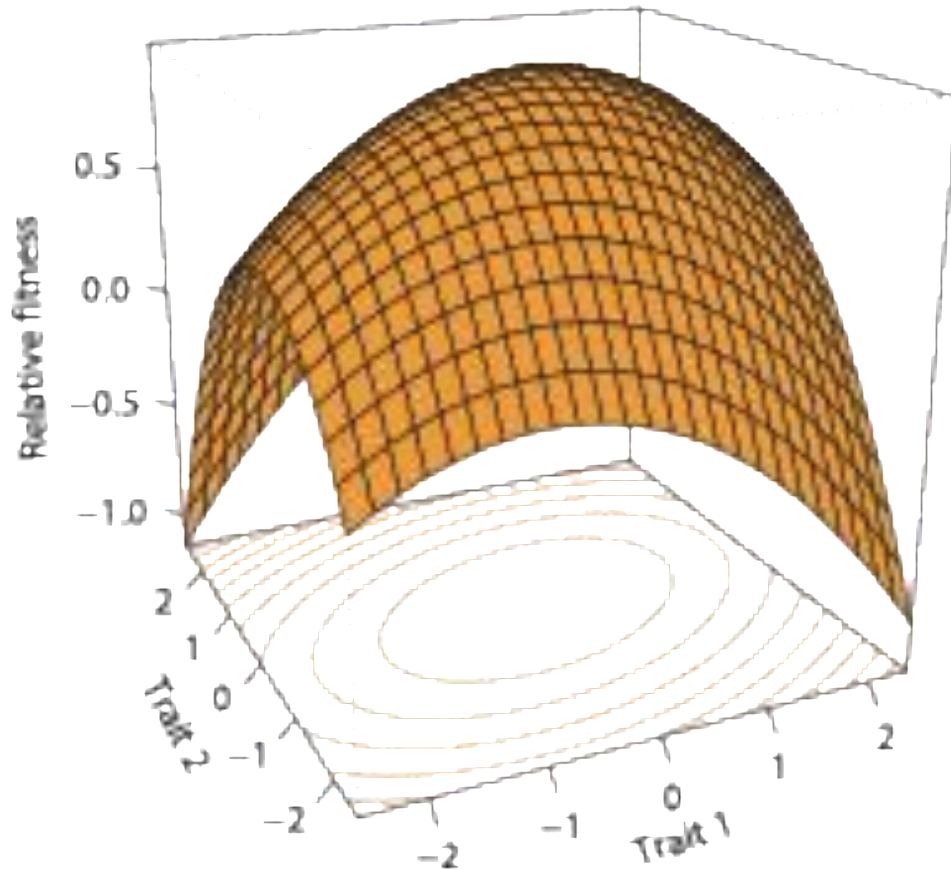


Figure 4.1 Bivariate stabilizing selection represented as a convex selection surface.

Seleção para aumentar MA:

Ótimo de força de mordida pode ser atingido por aumentar o out-lever e reduzir o in-lever OU reduzir o out-lever e aumentar o in-lever

Seleção estabilizadora em um contraste entre in-levers e outlever

Espécies de Tropiduridae com diferentes morfologias e ecologias



Tropidurus hispidus (Cerrado):

- Very territorial (microhabitat generalist)
- Non-flattened, large size



Tropidurus semitaeniatus (Caatinga):

- Flattened head (rock crevices)
- Medium size



Eurolophosaurus nanuzae (Cerrado):

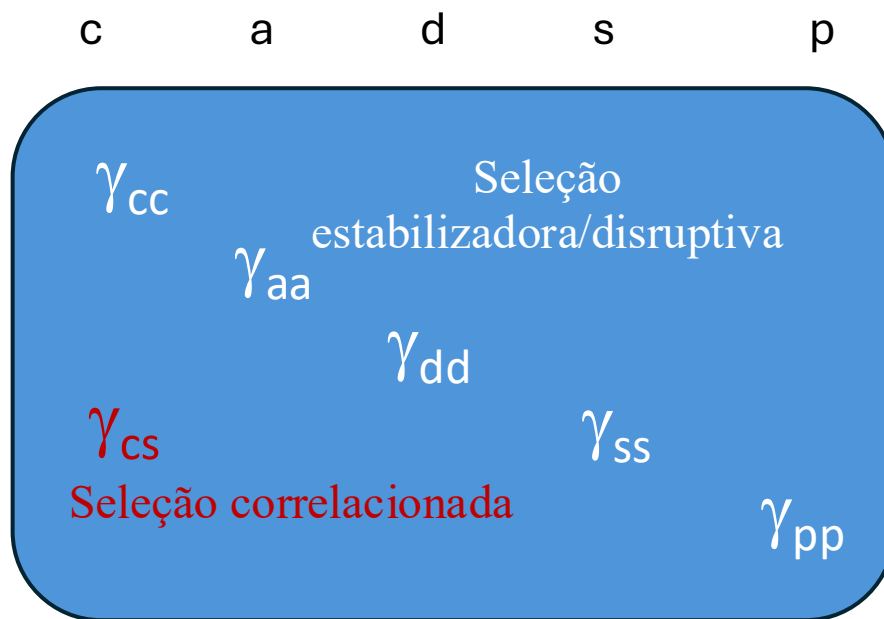
- Endemic to rocky meadows of MG/BA
- Non-flattened, small size



Estimativas de matriz de seleção estabilizadora multivariada

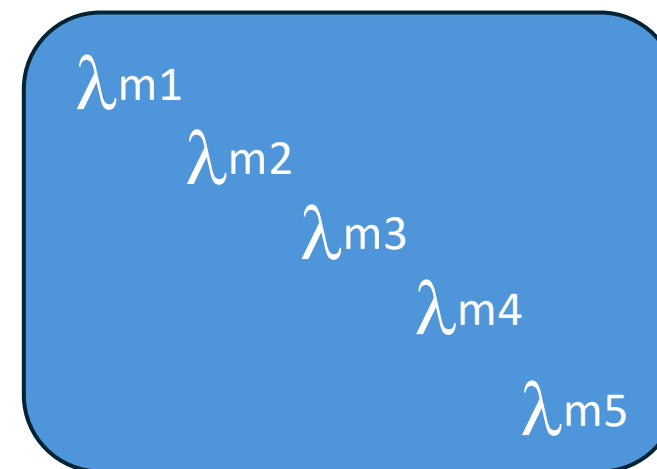
$$f = x_1\beta_1 + x_2\beta_2 + \frac{1}{2}x_1^2\gamma_1 + \frac{1}{2}x_2^2\gamma_2 + x_1x_2\gamma_{12} + cf.$$

coronoid, *c*
angular, *a*
dentary, *d*
surangular, *s*
prearticular, *p*



Análise Canônica (eigenanalysis)

$$\mathbf{\Lambda} = \mathbf{M}^T \mathbf{\gamma} \mathbf{M}$$

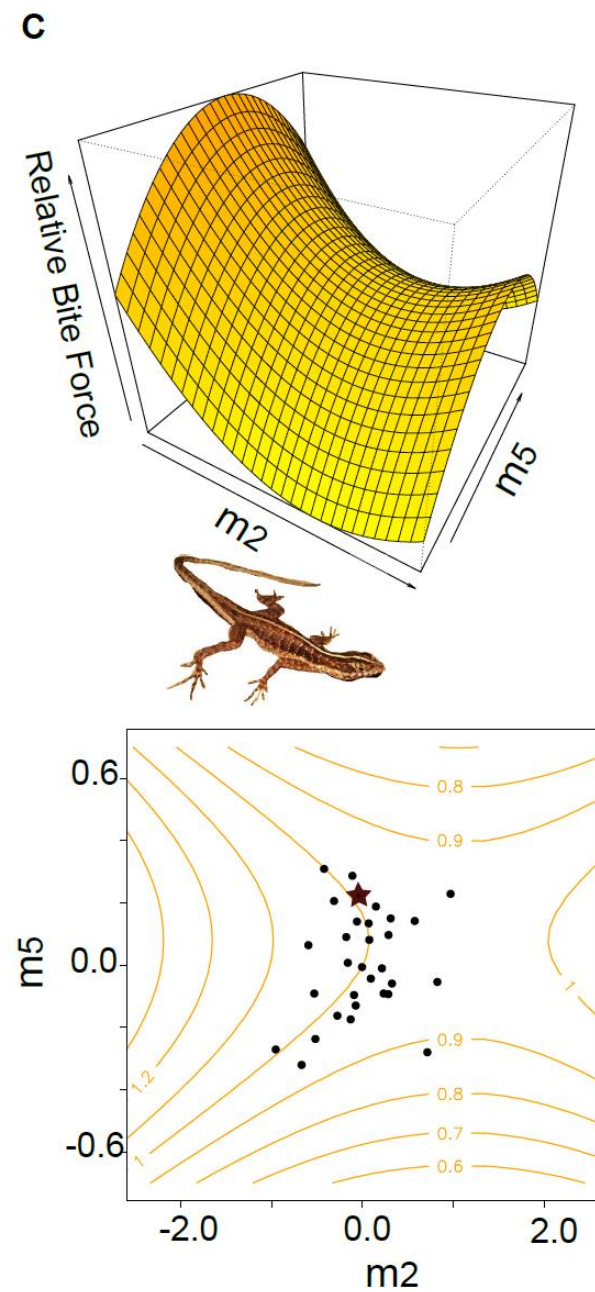
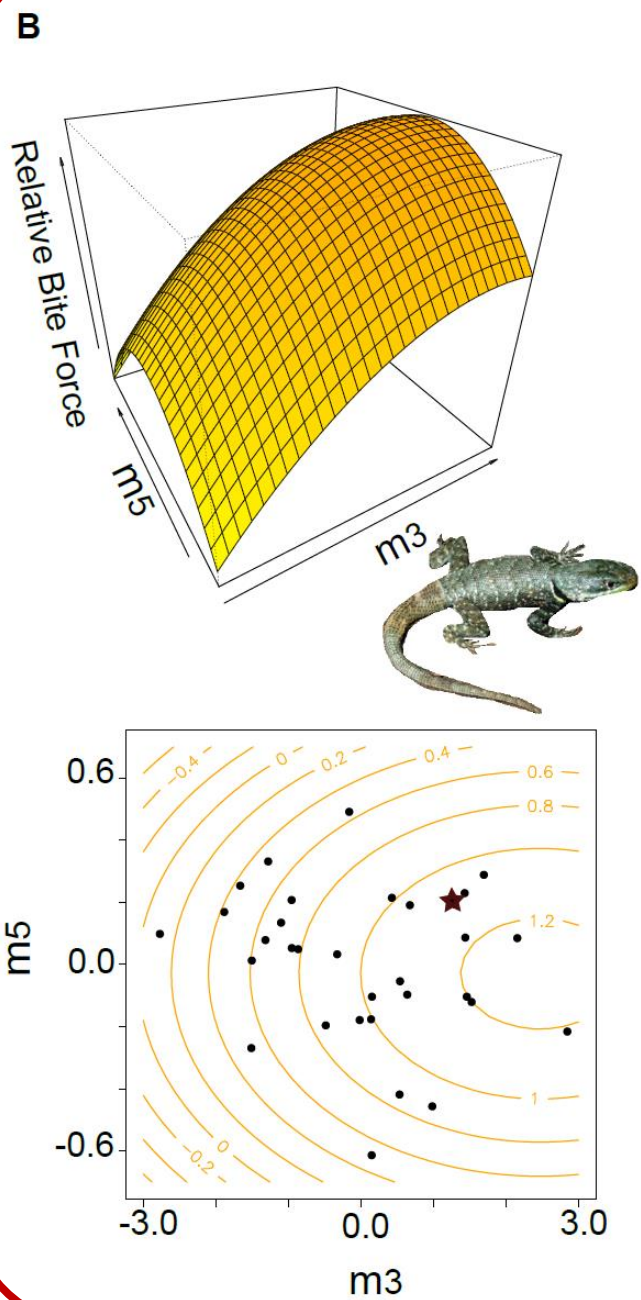
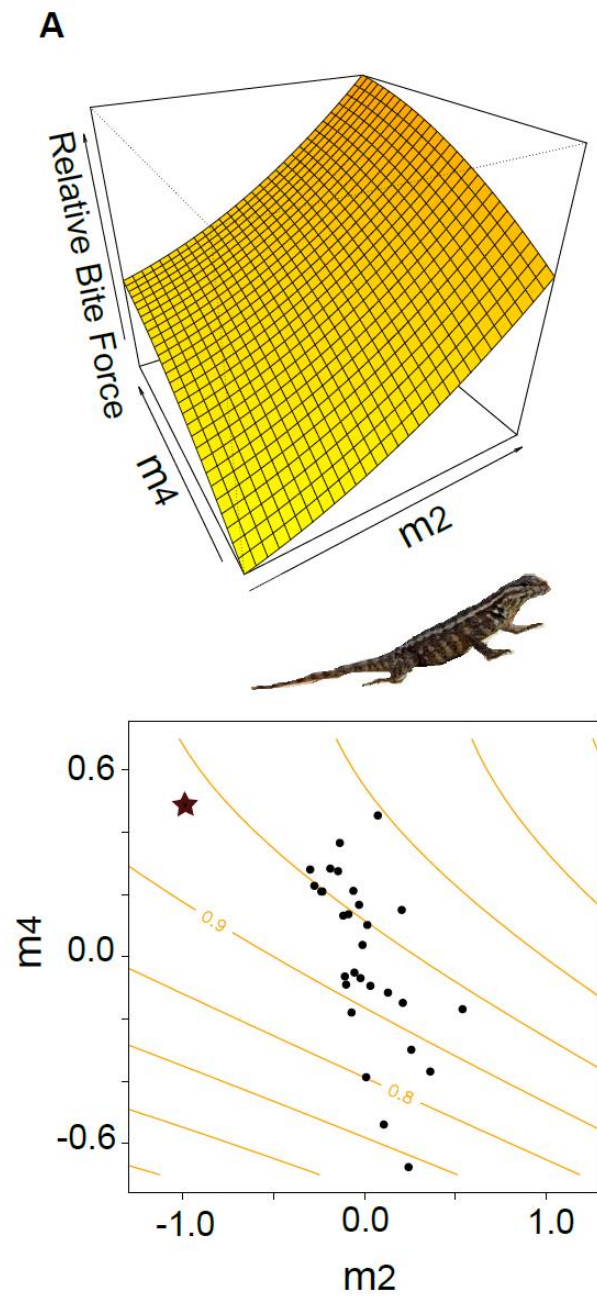


Autovetores (m1, m2, m3, m4, m5)
Autovalores (λ)

Previsão: autovalor negativo para um contraste entre in-lever e out-lever

Traits, abbreviations	Eigenvectors of γ				
	m_1	m_2	m_3	m_4	m_5
coronoid, <i>c</i>	0.851	-0.463	0.113	-0.103	-0.194
angular, <i>a</i>	0.066	0.403	0.179	-0.890	-0.095
dentary, <i>d</i>	0.035	0.149	0.935	0.235	0.219
surangular, <i>s</i>	-0.504	-0.759	0.242	-0.323	-0.088
prearticular, <i>p</i>	-0.126	0.159	0.152	0.195	-0.947
Eigenvalue (λ_i)	5.997 ns	2.468 ns	-0.162 ns	-3.263 ns	-6.104*
In-lever mapping	IN_1	IN_2	no	no	IN_3

* denotes $P = 0.07$, ns denotes $P > 0.17$. Highlighted cells indicate the expected participation of bony elements in in-levers.

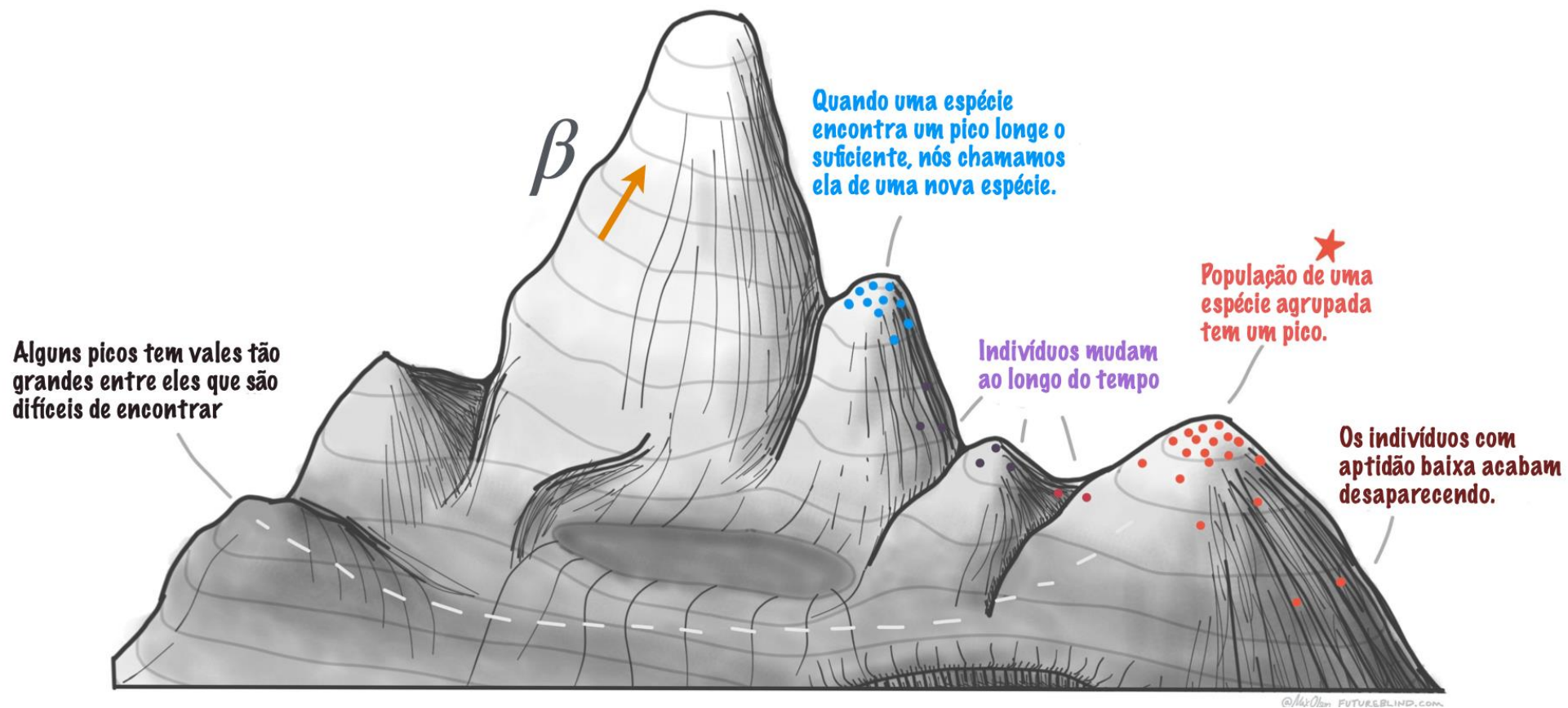




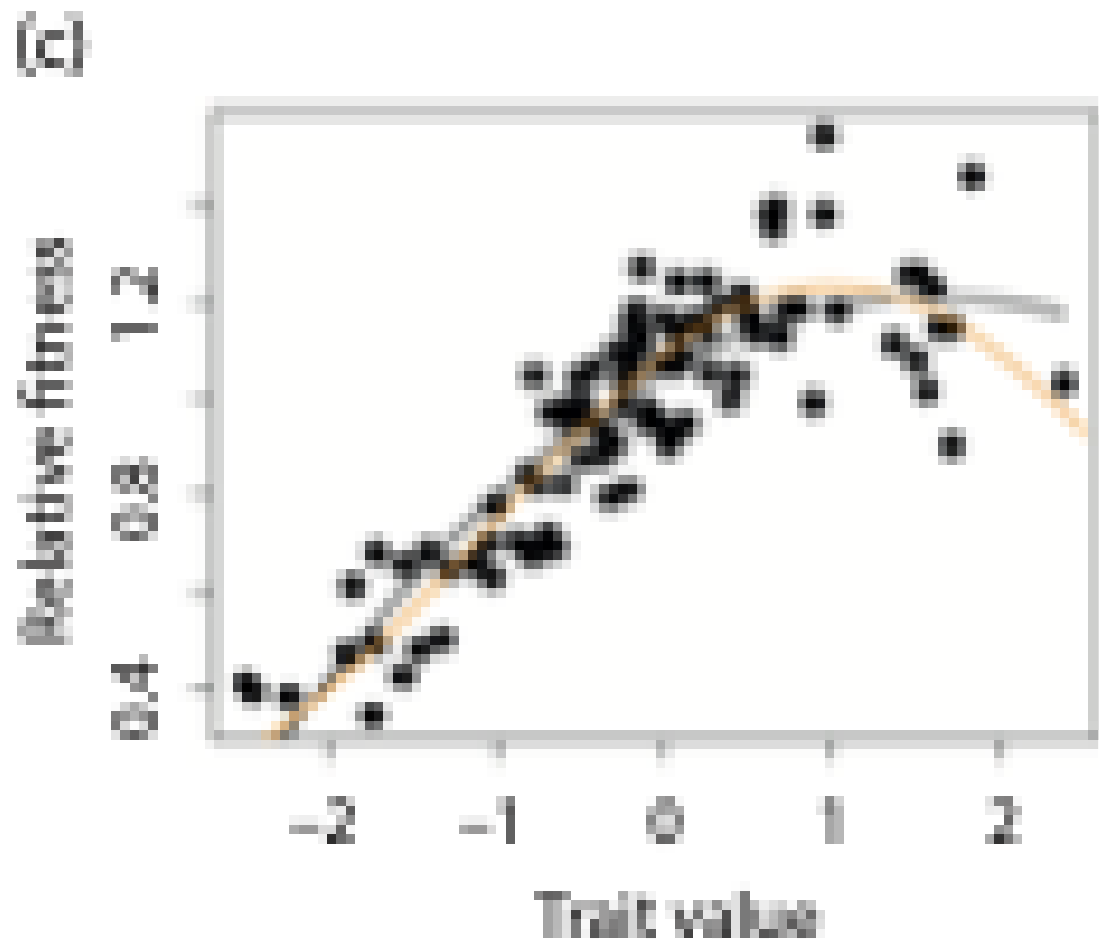
Oregon State
University



E a Paisagem Adaptativa??



Função Gaussiana e ISS ‘verdadeira’



ÓTIMO
ADAPTATIVO!

$$w(z) = \exp \left\{ \frac{-(z - \theta)^2}{2\sigma^2} \right\},$$

abertura da função

ISS = aptidão relativa dos indivíduos em função de seus valores de z

ISS multivariada e Paisagem Adaptativa (AL)

AL = aptidão média populacional em função dos valores médios de $\mathbf{z}$

ISS = Gaussiana multivariada

$$w(\mathbf{z}) = \exp \left\{ -\frac{1}{2} (\mathbf{z} - \boldsymbol{\theta})^T \boldsymbol{\omega}^{-1} (\mathbf{z} - \boldsymbol{\theta}) \right\}$$

AL = Média da Gaussiana multivariada sobre a distribuição multivariada dos caracteres

$$\ln \bar{W} = \exp \left\{ -\frac{1}{2} (\bar{\mathbf{z}} - \boldsymbol{\theta})^T \boldsymbol{\Omega}^{-1} (\bar{\mathbf{z}} - \boldsymbol{\theta}) \right\}$$

Matriz de forma e orientação da paisagem

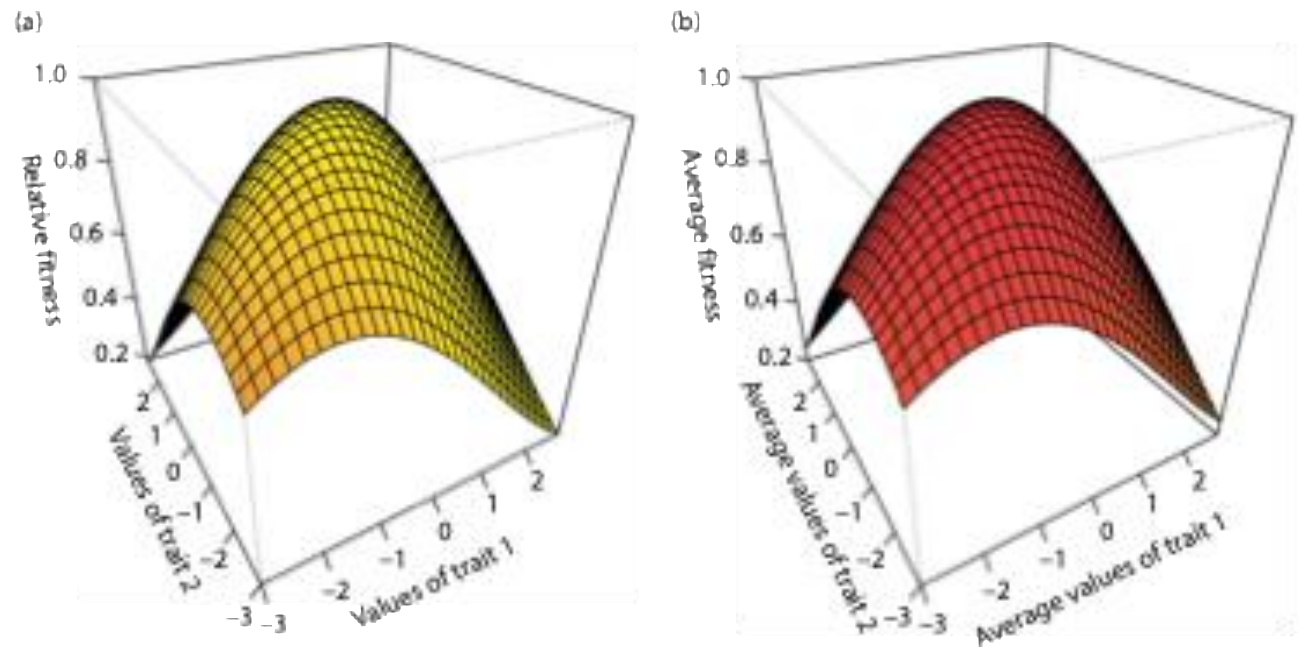
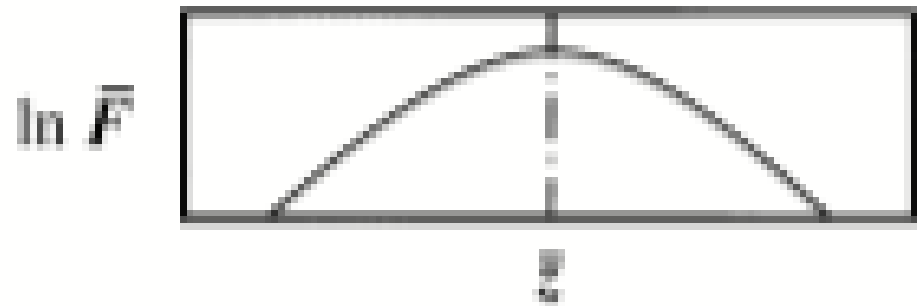


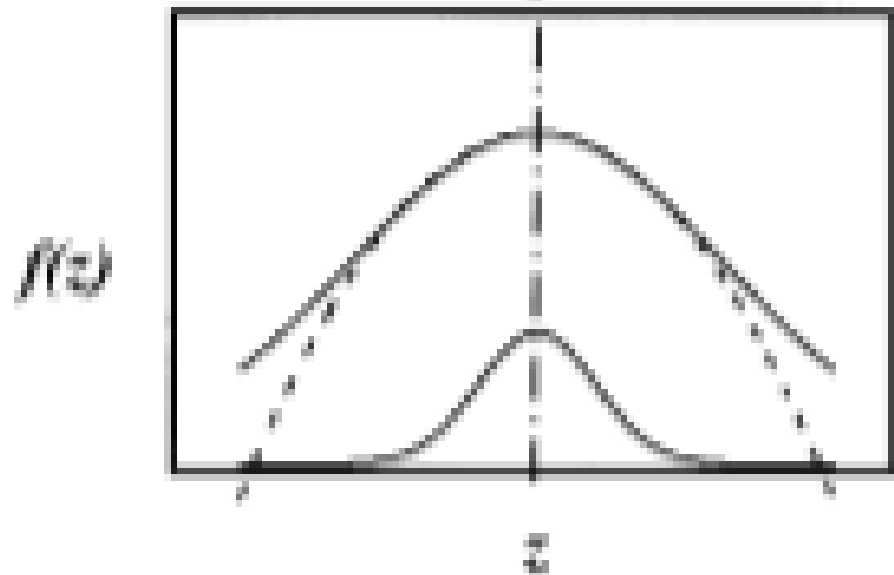
Figure 4.6 A Gaussian ISS (a) and corresponding Gaussian AL (b) are similar in configuration if stabilizing selection is weak, as in this example: Gaussian ISS with $\boldsymbol{\theta} = (0 \ 0)^T$ and $\boldsymbol{\omega} = \begin{pmatrix} 5 & 2.5 \\ 2.5 & 5 \end{pmatrix}$ and the corresponding Gaussian AL with $\boldsymbol{\Omega} = \begin{pmatrix} 6 & 2.5 \\ 2.5 & 6 \end{pmatrix}$, where $\mathbf{P} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

Paisagem de Desempenho



β_f : direção de maior inclinação da paisagem

$\gamma = \beta\beta^T$: curvatura da paisagem



$p(z)$ Aproximação quadrática da superfície de desempenho

$$f = z_1\beta_{f1} + z_2\beta_{f2} + \frac{1}{2}z_1^2\gamma_{f11} + \frac{1}{2}z_2^2\gamma_{f22} + z_1z_2\gamma_{f12} + cr.$$

Relações entre superfícies e paisagens

TABLE 1. *Nomenclature for surfaces and landscapes*

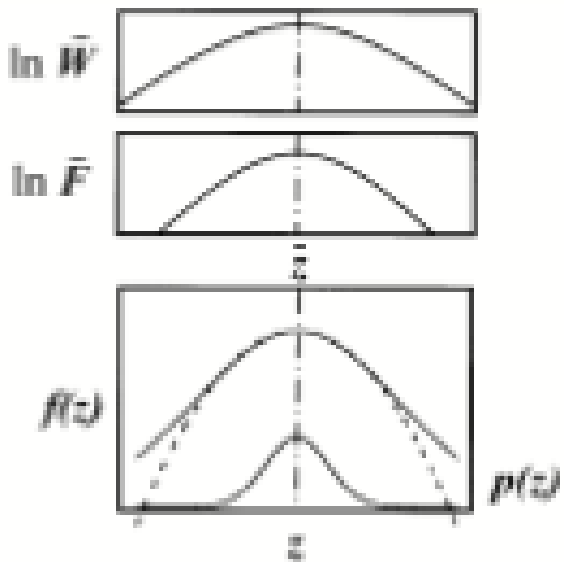
Name of surface or landscape	Dimensions			
	Vert.	Hori.	Slope ¹	Curvature ²
Individual selection surface (selection surface)	$w(z)$	z	β	γ
Adaptive landscape for morphology (adaptive landscape)	$\ln \bar{W}$	$\bar{z}$	β	$\gamma - \beta\beta^T$
Individual performance surface (performance surface)	$f(z)$	z	β_f	γ_f
Performance landscape for traits (performance landscape)	$\ln \bar{F}$	$\bar{z}$	β_f	$\gamma_f - \beta_f\beta_f^T$
Individual selection surface for performance (selection surface for performance)	$w(f)$	f	β_w	γ_w
Adaptive landscape for performance (performance adaptive landscape)	$\ln \bar{W}$	f	β_w	$\gamma_w - \beta_w\beta_w^T$

¹ The symbol represents the average slope in the case of individual surfaces.

² The symbol represents the average curvature in the case of individual surface.

$$f(z) = \exp \left[-\frac{1}{2}(z - \theta_f)^T \omega_f^{-1} (z - \theta_f) \right]$$

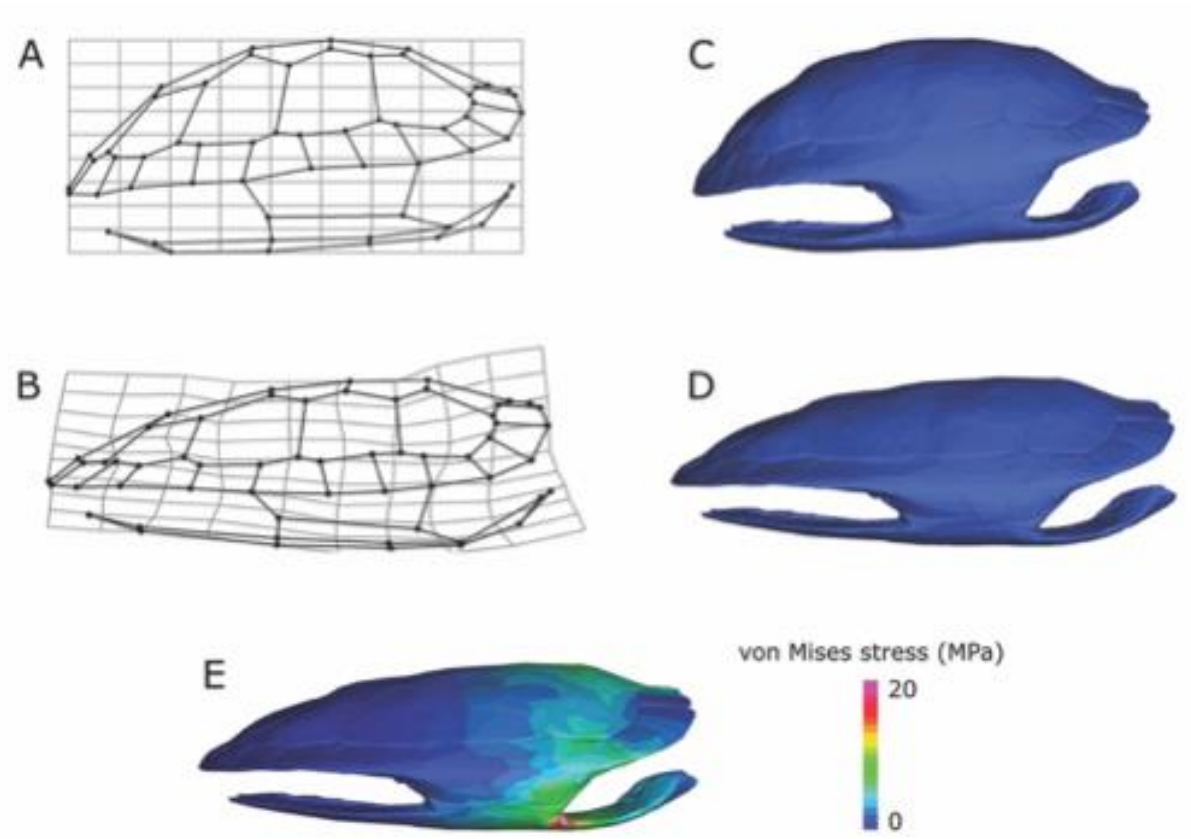
ÓTIMO OU PICO DE DESEMPENHO!



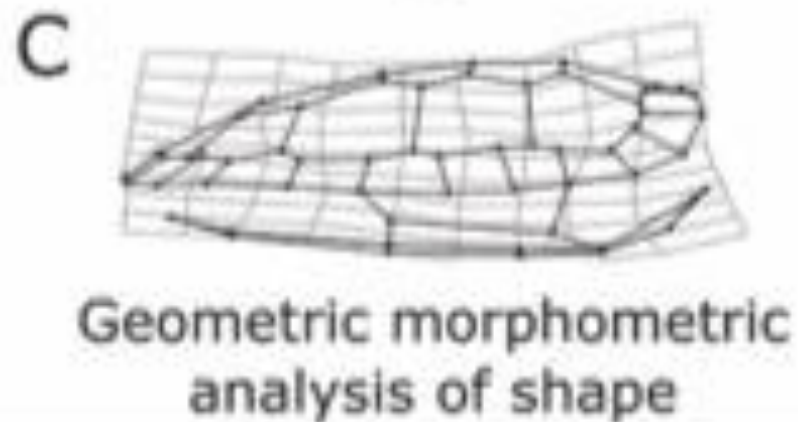
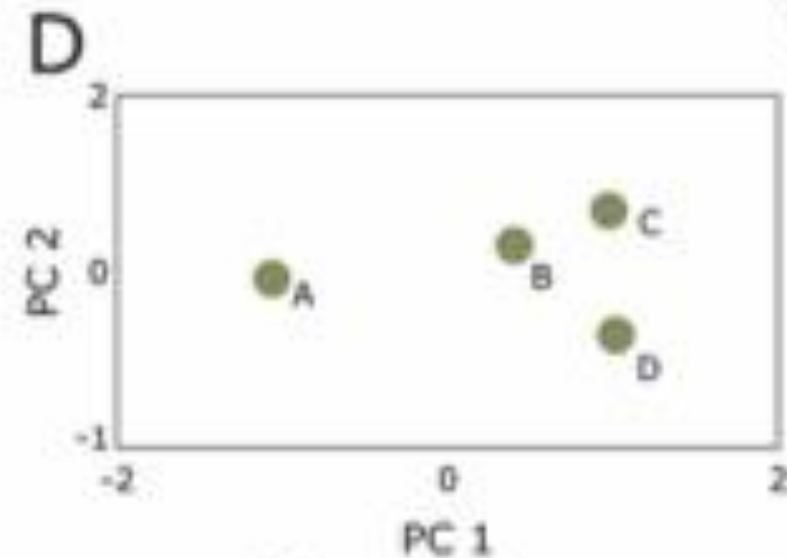
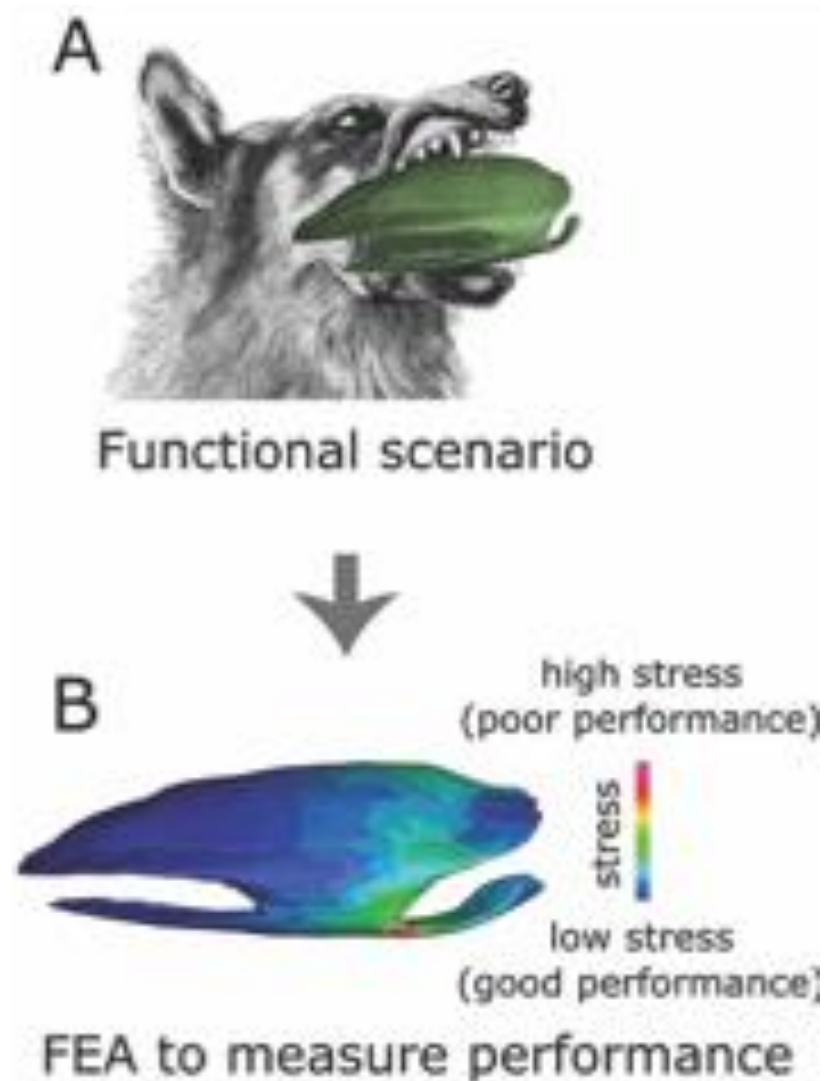
Morfometria geométrica e Paisagens de desempenho

COMBINING GEOMETRIC MORPHOMETRICS AND FINITE ELEMENT ANALYSIS WITH EVOLUTIONARY MODELING: TOWARDS A SYNTHESIS

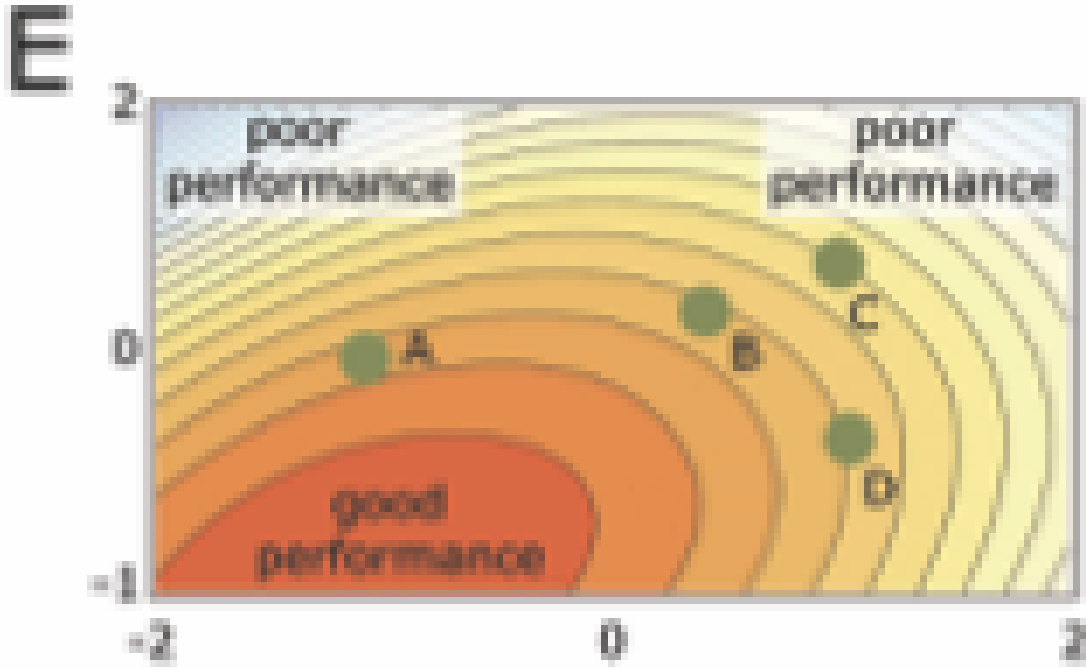
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EMILY J. RAYFIELD,⁵ and KENNETH D. ANGIELCZYK^{*,6}



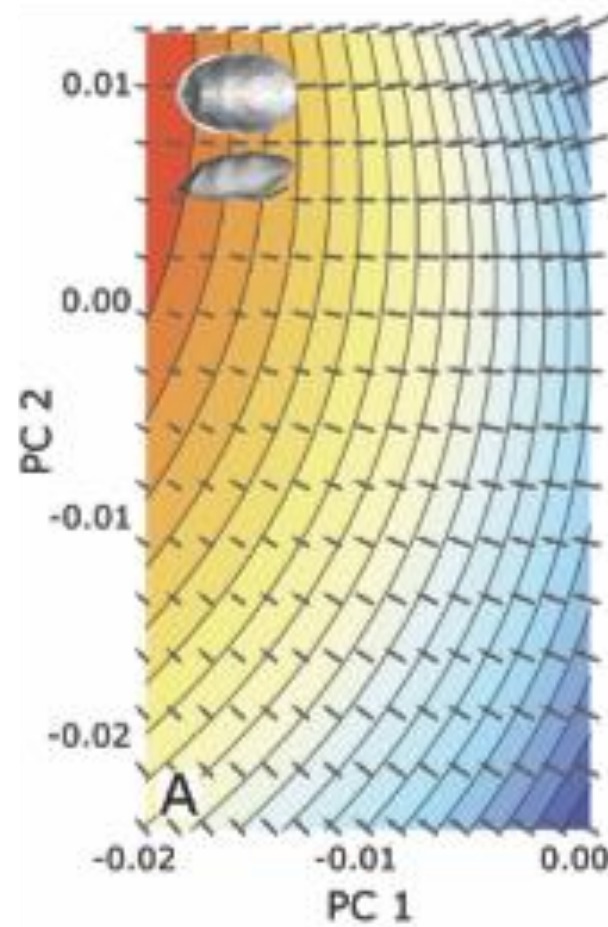
Morfometria geométrica



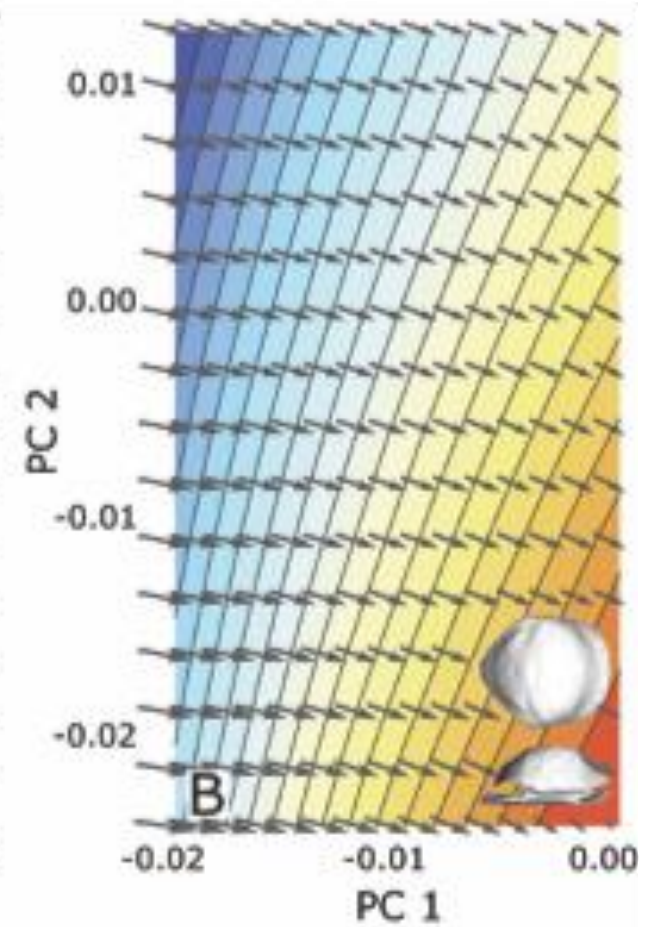
PCs de Forma e Paisagens de desempenho



Aproximação quadrática da paisagem de desempenho



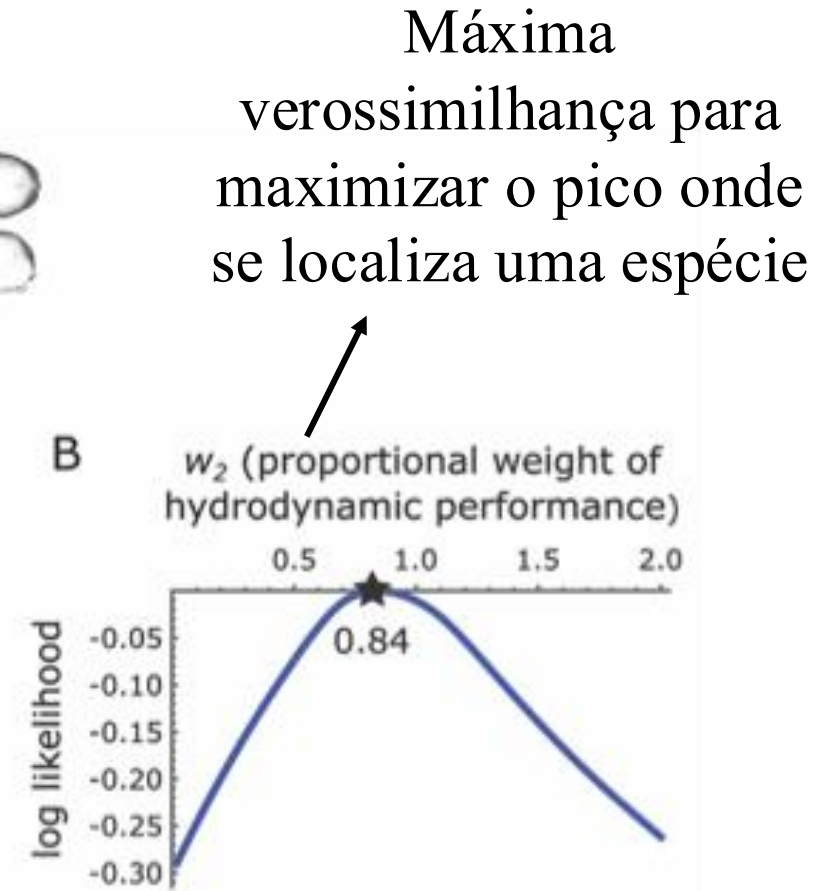
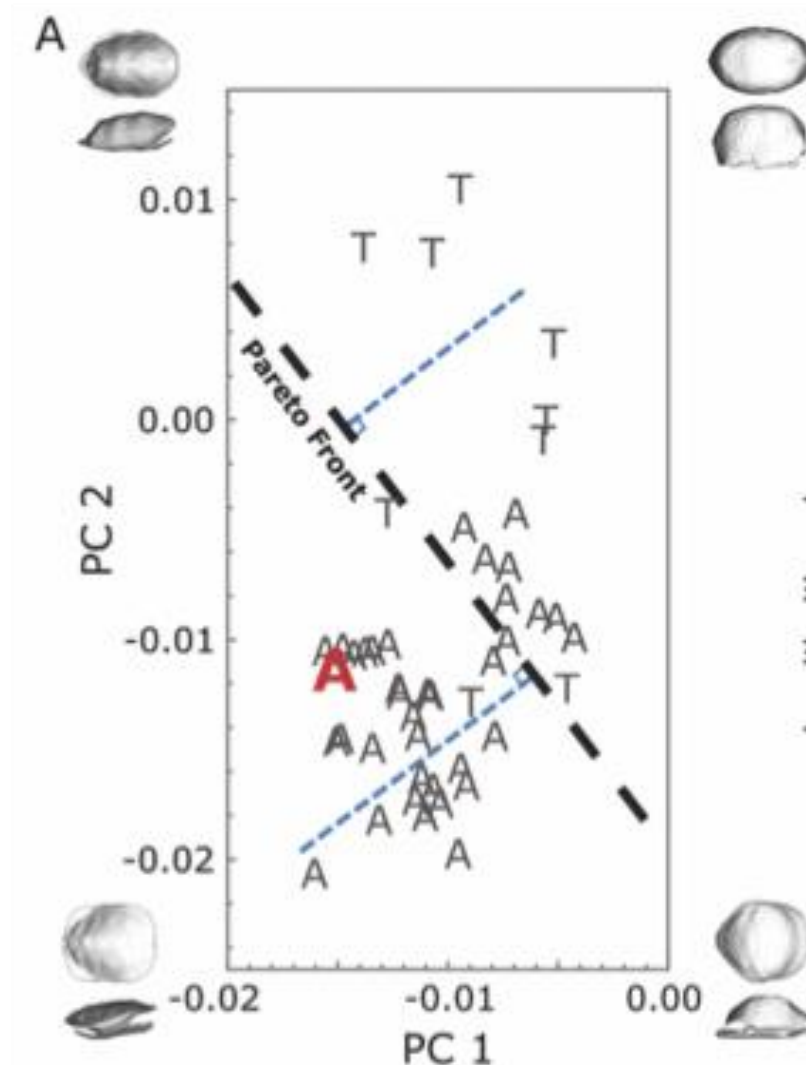
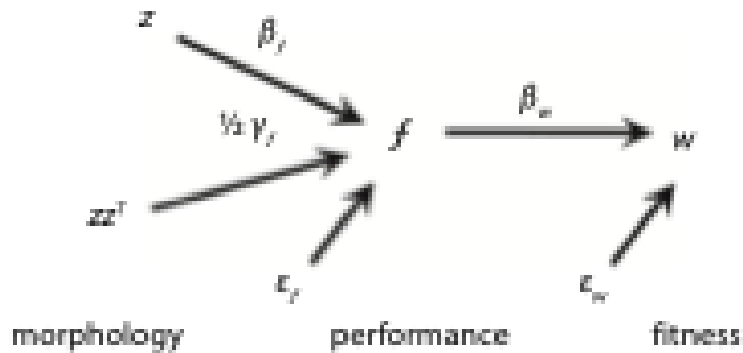
Resistência ao estresse



Hidrodinâmica

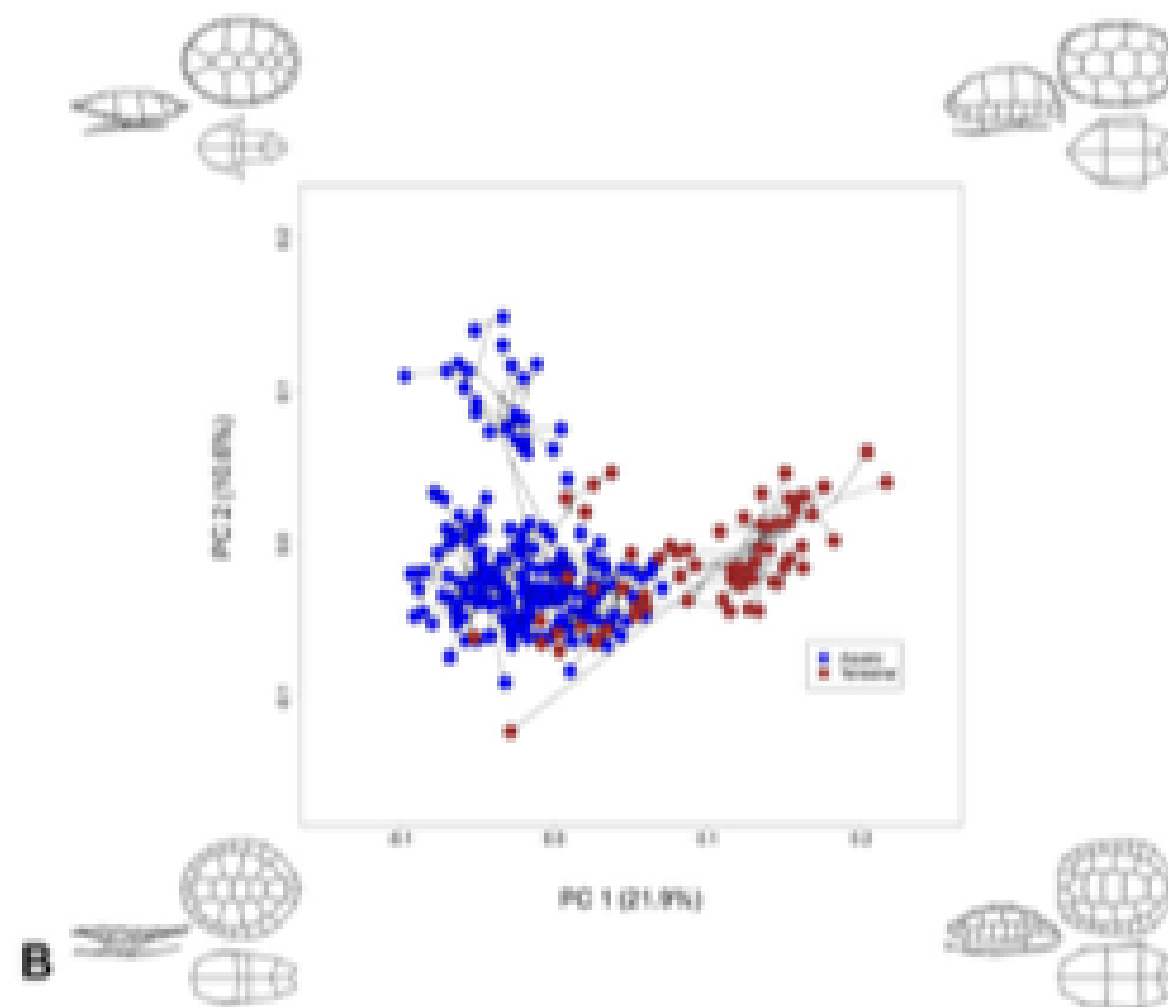
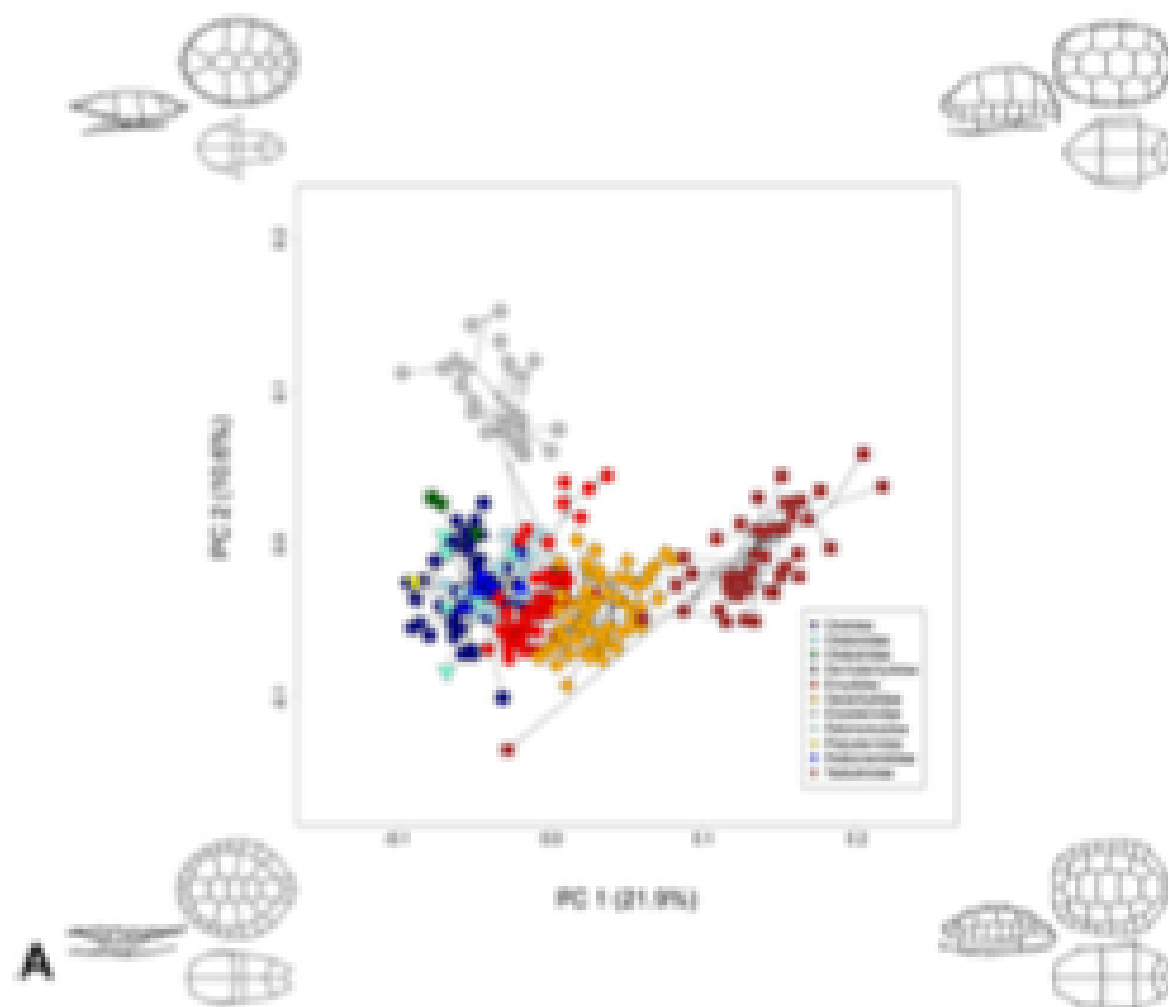
O 'pulo' para Paisagens Adaptativas: o quanto o desempenho afeta aptidão média?

$$\bar{W} = w_1 F_1 + w_2 F_2 + \dots$$



Performance in three shell functions predicts the phenotypic distribution of hard-shelled turtles

C. Tristan Stayton^{1,2} 

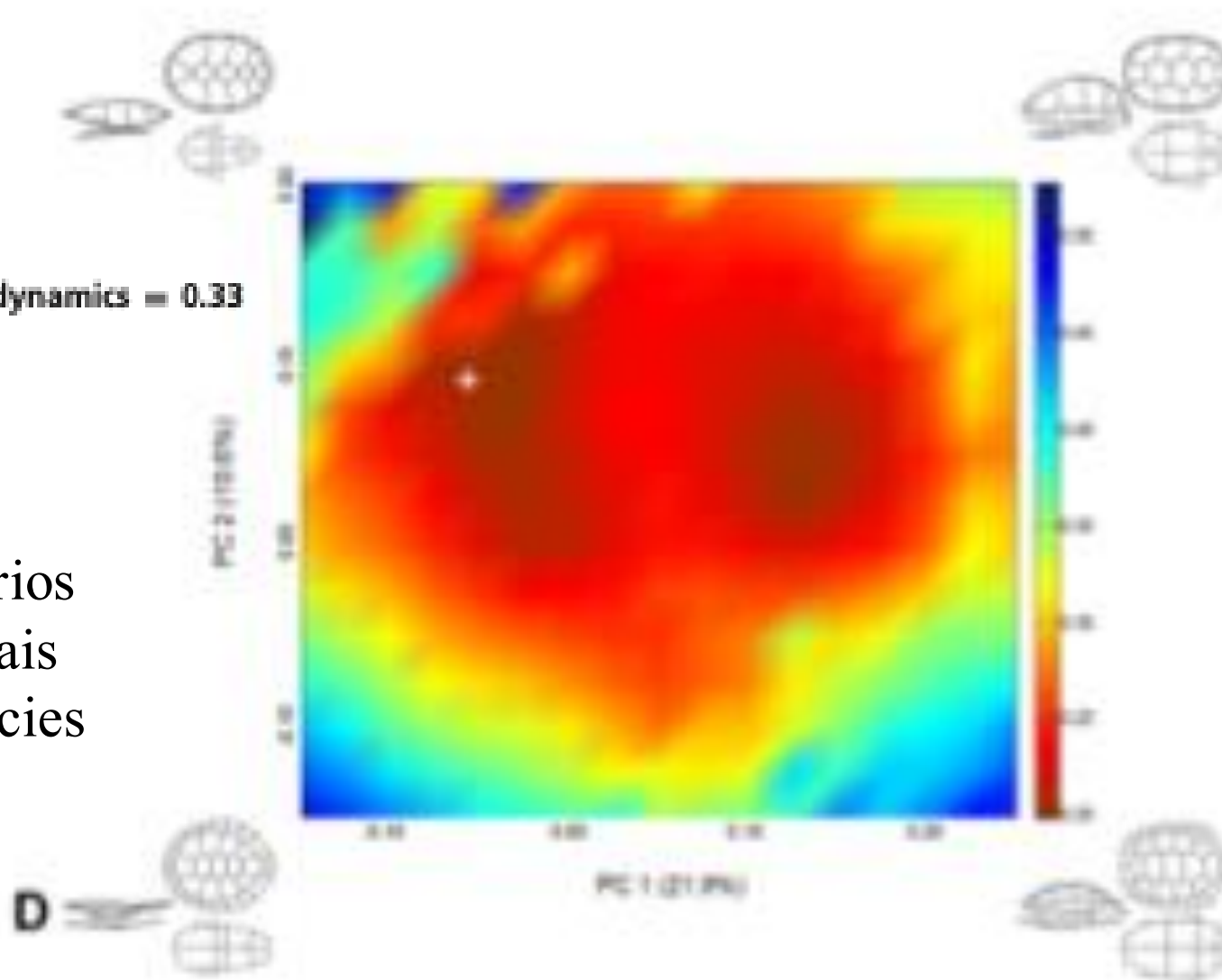


Paisagem de Desempenho Combinada

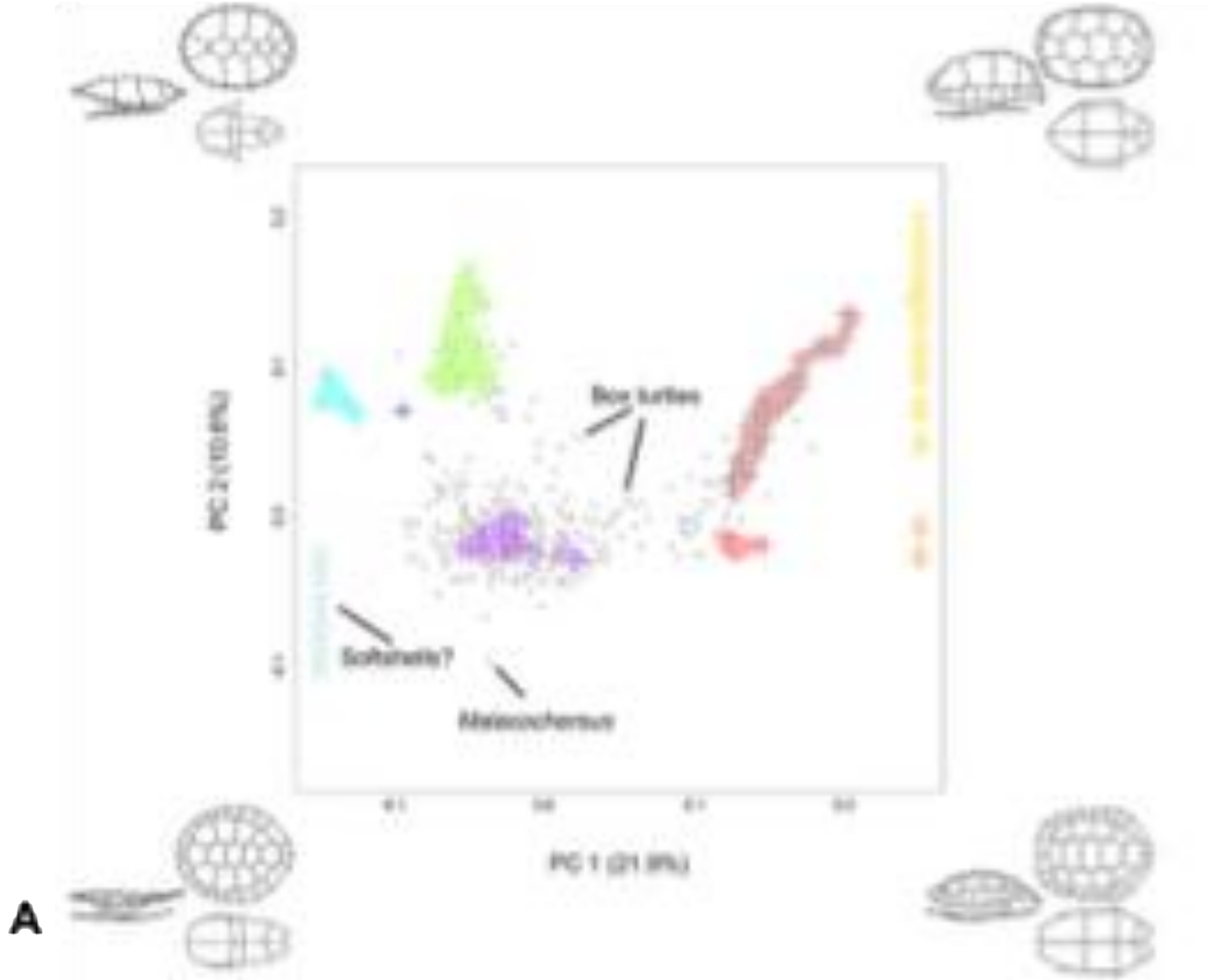
strength = 0.34, for hydrodynamics = 0.33
righting = 0.33



Simulação de vários
pesos – picos mais
próximos às espécies



Picos de Desempenho refletem Picos Adaptativos?



Bridging Performance and Adaptive Landscapes to Understand Long-Term Functional Evolution

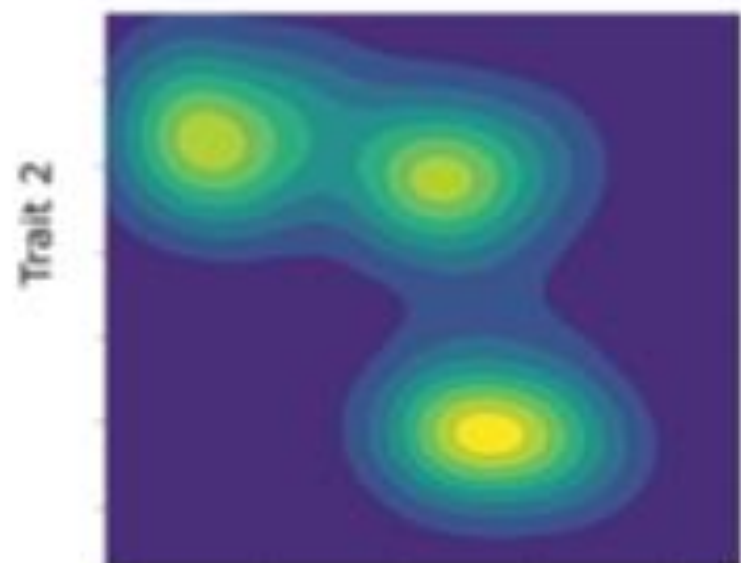
Monique Nouailhetas Simon*

Daniel S. Moen

Hypothetical phenotypes + Functional models

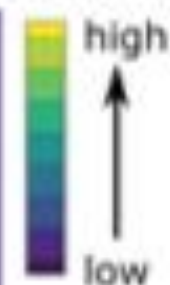


Performance Landscape



Trait 1

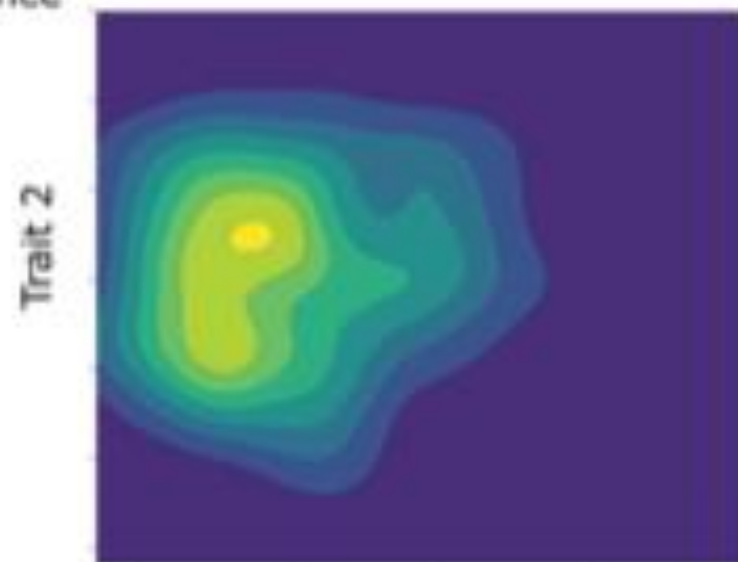
Performance



Species phenotypes + Phylogeny + OU models

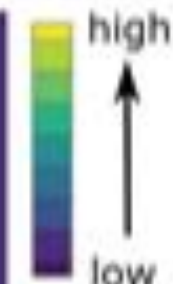


Adaptive Landscape



Trait 1

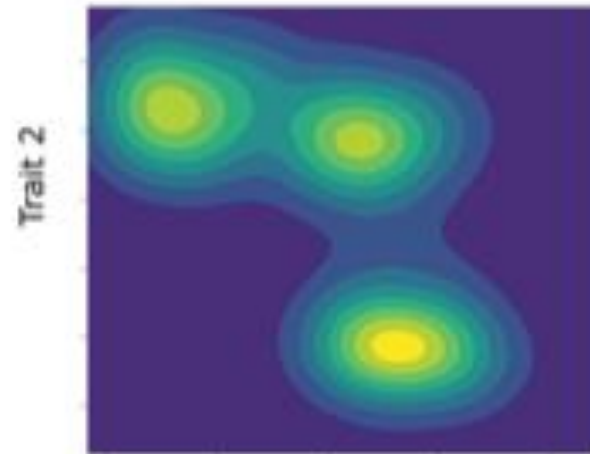
Fitness



Hypothetical phenotypes + Functional models

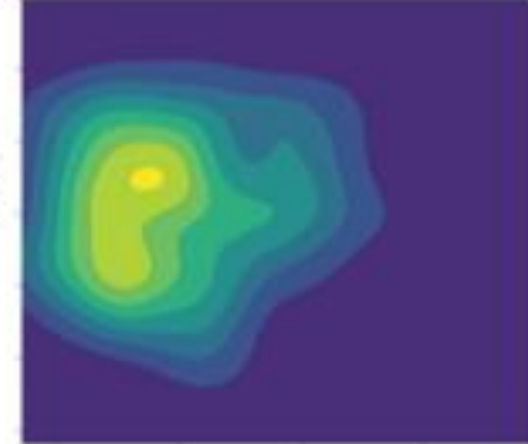
Species phenotypes + **Phylogeny** + OU models

Performance Landscape



Performance
high
low

Adaptive Landscape



Fitness
high
low

Trait 1

Trait 1

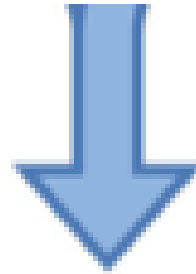
Direct link between phenotypic variation
and function fitted to different ecologies

Direct link between phenotypic variation
and fitness fitted to different ecologies

+
Evolutionary constraints

Merged Framework

Merged Framework



QUESTION: Do phenotypes that maximize performance also maximize fitness?

APPROACH: Compare the locations of performance and adaptive peaks

QUESTION: How widespread are constraints on functional adaptation?

APPROACH: Compare species phenotypes with inferred primary adaptive peaks

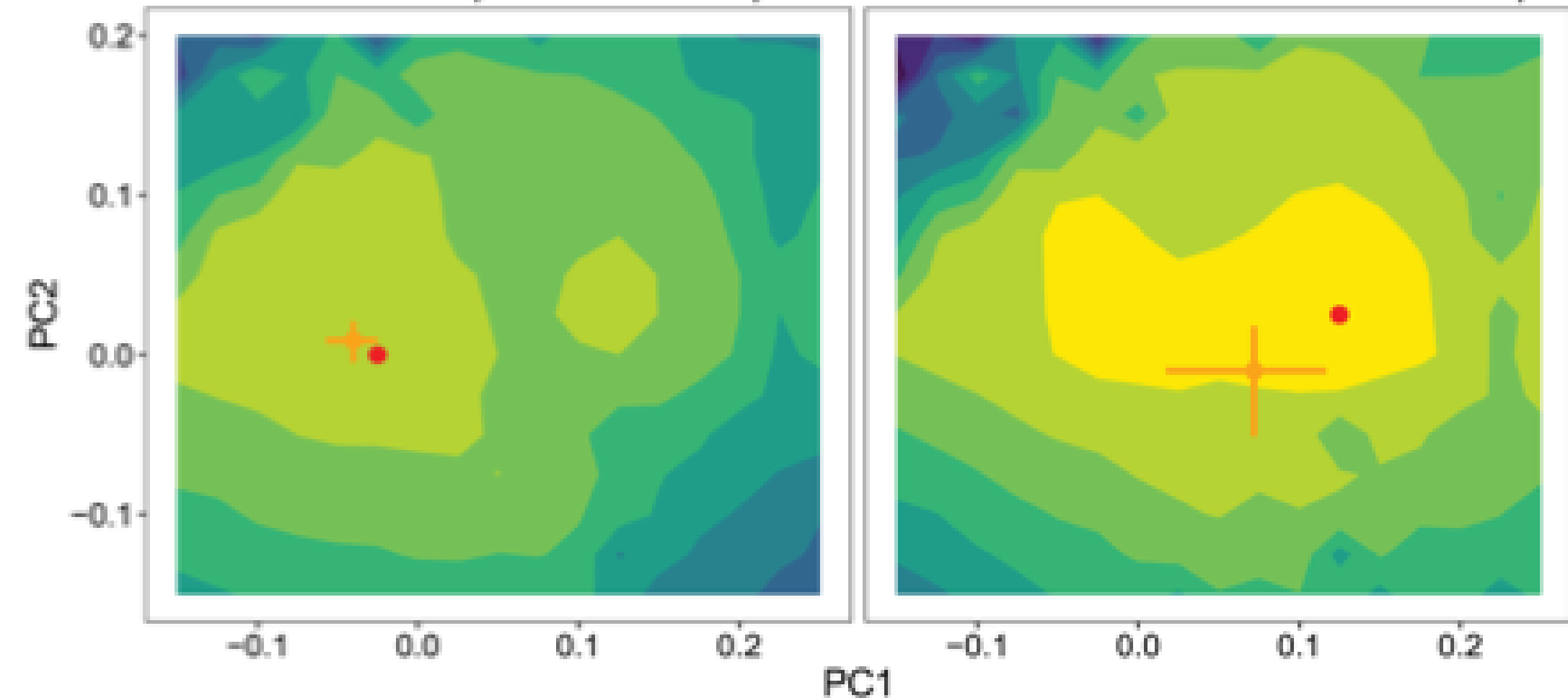
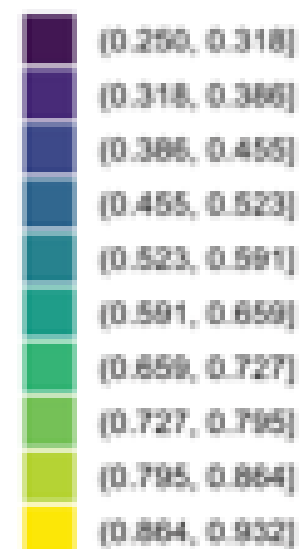
A. Combined aquatic landscape

B. Combined terrestrial landscape

Peak type

- ALA
- PLA

Performance



aquatic landscape: shell strength = 0.47, hydrodynamic drag = 0.38, self-righting ability = 0.15;

terrestrial landscape: shell strength = 0.68, hydrodynamic drag = 0.18, self-righting ability = 0.16